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AI Assisted Epilepsy detection using EEG signals

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Abstract

Epilepsy is a prevalent neurological disorder affecting approximately 50 million people worldwide. Automated analysis of electroencephalogram (EEG) signals may help reduce the burden of manual seizure review. This study proposes a hybrid deep learning architecture combining a 1D Convolutional Neural Network (1D-CNN), Bidirectional Long Short-Term Memory network (Bi-LSTM), and Self-Attention mechanism for automated epileptic seizure detection using the University of Bonn EEG dataset. A strict recording-level 10-Fold cross-validation strategy is employed to prevent data leakage from overlapping sliding-window segments. The proposed model is systematically compared against three baselines — SVM with wavelet-packet features, a pure 1D-CNN, and a pure Bi-LSTM with attention — in an ablation study across binary, three-class, and five-class classification tasks. Experimental results show that the hybrid model achieves 99.67% accuracy on binary classification, 97.69% on three-class, and 84.51% on the five-class task. On the five-class task, it shows statistically significant gains over the pure CNN ($p = 0.001$) and pure Bi-LSTM ($p = 0.001$) baselines, while the mean gain over SVM does not reach significance at the 0.05 level. Attention weight visualisation highlights salient temporal regions and provides a qualitative view of model behaviour.

Keywords

Epilepsy Detection, Electroencephalogram (EEG), Deep Learning, 1D Convolutional Neural Network, Bidirectional LSTM, Self-Attention, Bonn Dataset, Cross-Validation

摘要

癫痫是一种常见的神经系统疾病，全球约有 5000 万患者。脑电图（EEG）信号的自动分析为辅助临床医生进行癫痫发作检测提供了一种有前景的方法。本研究提出了一种结合一维卷积神经网络（1D-CNN）、双向长短期记忆网络（Bi-LSTM）和自注意力机制的混合深度学习架构，利用波恩大学 EEG 数据集实现癫痫发作的自动检测。本研究采用严格的录音级别 10 折交叉验证策略，以防止滑动窗口重叠导致的数据泄露。所提出的模型与三个基线模型——基于小波包特征的 SVM、纯 1D-CNN 和纯 Bi-LSTM——在二分类、三分类和五分类任务上进行了系统性消融实验对比。实验结果表明，混合模型在二分类任务上达到 99.67% 的准确率，三分类任务达到 97.69%，五分类任务达到 84.51%，在五分类任务上显著优于纯 CNN ($p = 0.001$) 和纯 Bi-LSTM ($p = 0.001$) 基线模型。注意力权重可视化验证了模型能够聚焦于具有临床意义的时间特征，增强了模型的可解释性。

关键词

癫痫检测，脑电信号，深度学习，一维卷积神经网络，双向长短期记忆网络，自注意力机制，波恩数据集，交叉验证

Chapter 1: Introduction

1.1 Research Background and Motivation

Epilepsy affects roughly 50 million people worldwide. The condition causes recurrent seizures — episodes of abnormal, excessive electrical activity in the brain — and carries consequences well beyond the seizures themselves: cognitive decline, social stigma, and a real risk of injury during an episode. In developing countries, where access to neurological specialists is limited, the diagnostic gap is even wider; up to 75% of epilepsy patients in low-income regions do not receive appropriate treatment.

Clinicians diagnose epilepsy primarily through the electroencephalogram (EEG), which records brain electrical activity via scalp or intracranial electrodes. A routine clinical workup may involve 24 to 72 hours of continuous EEG monitoring, all of which a neurologist must visually inspect to identify seizure events (ictal activity) and inter-ictal epileptiform discharges. This manual review is slow, subjective, and dependent on specialist expertise that is simply not available in many healthcare settings. Even among trained neurologists, inter-rater agreement on subtle patterns — such as non-convulsive seizures and brief inter-ictal discharges — can be surprisingly low.

Automated EEG analysis has long been pursued as a way to address these problems. Early work relied on handcrafted features (wavelet coefficients, spectral power, statistical descriptors) fed into classifiers such as Support Vector Machines. These pipelines worked reasonably well for binary seizure detection, but they depend heavily on the choice of features and the expertise of whoever designs them. When the task becomes harder — distinguishing between different types of abnormal brain activity, for instance — handcrafted features often fall short.

Recent deep learning methods have made it possible to learn EEG features directly from the signal itself. Convolutional Neural Networks (CNNs) can capture local morphological patterns — spike-wave complexes and sharp transients — without manual feature engineering. Recurrent networks, particularly Long Short-Term Memory (LSTM) architectures, model how those patterns evolve over time, while attention mechanisms allow the model to assign greater weight to the more informative parts of a recording.

However, several issues still persist. Some published studies report very high accuracy under protocols that do not clearly separate segments from the same recording across training and validation. Many Bonn evaluations also focus on binary classification (normal vs. seizure), which is usually easier than finer-grained multi-class discrimination because the signal differences are large. Controlled ablation studies are less common, so it is often unclear which architectural component is responsible for the reported gain.

This project addresses these gaps by combining CNN, Bi-LSTM, and Self-Attention into a single

hybrid architecture, evaluated under a strict recording-level cross-validation protocol on the University of Bonn EEG dataset across three tasks of increasing difficulty.

1.2 Research Aims and Objectives

The aim of this project is to build and evaluate a hybrid deep learning model — specifically, a 1D-CNN combined with a Bidirectional LSTM and Self-Attention mechanism — for automated epilepsy seizure detection from EEG signals.

Five objectives guide the work:

1. Implement a strict recording-level cross-validation framework on the University of Bonn EEG dataset, preventing the data leakage that undermines many published results in this area.
2. Design and implement the hybrid 1D-CNN + Bi-LSTM + Self-Attention architecture, integrating local feature extraction, temporal modelling, and attention weighting in a single end-to-end pipeline.
3. Build three baseline models — SVM with wavelet-packet features, a pure 1D-CNN, and a pure Bi-LSTM with attention — to run as an ablation study and isolate the contribution of each component.
4. Evaluate all four models across three classification tasks (binary, three-class, five-class) with paired t-tests to check whether performance differences are statistically meaningful.
5. Visualise the learned attention weights to show which temporal regions of the EEG the model relies on, providing a qualitative view of model behaviour.

1.3 Report Structure

The remainder of this report is organised as follows.

Chapter 2 reviews the clinical and technical background: EEG signals and their role in epilepsy diagnosis, existing machine learning and deep learning approaches to seizure detection, attention mechanisms in biomedical signal processing, and the research gaps this project addresses.

Chapter 3 describes the Bonn EEG dataset, the preprocessing pipeline including the recording-level splitting strategy and the data leakage issue that was discovered and corrected, the proposed hybrid model architecture, the three baseline models, and the experimental setup.

Chapter 4 presents classification results across all three tasks, analyses error patterns through confusion matrices, visualises attention weights for model interpretability, conducts statistical

significance testing, compares results with published literature, and discusses clinical applicability and limitations.

Chapter 5 concludes the report with a summary of key findings, reflections on the methodology and lessons learned, and directions for future work.

Chapter 2: Background and Related Work

2.1 EEG Signals and Epilepsy Clinical Diagnosis

The brain generates electrical signals through the coordinated firing of millions of neurons. An EEG captures these signals — typically at sampling rates between 128 and 512 Hz — using electrodes placed on the scalp (surface EEG) or directly on the brain surface (intracranial EEG). The resulting waveforms are usually described in terms of frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (above 30 Hz).

For epilepsy diagnosis, three types of EEG patterns matter. *Ictal* activity, recorded during a seizure, shows high-amplitude rhythmic discharges (often exceeding 1000 μV) caused by synchronised neuronal firing. *Inter-ictal* activity, recorded between seizures, may contain epileptiform discharges — sharp waves and spike-wave complexes — that help localise the epileptogenic zone. *Normal* activity shows typical physiological rhythms with no pathological transients, usually below 200 μV .

In practice, a neurologist reviews hours (sometimes days) of continuous EEG to spot these patterns. The process is tedious, and different clinicians do not always agree on what they see. This is especially true for subtle inter-ictal patterns and non-convulsive seizures. The result is a clear motivation for automated analysis tools.

2.2 Traditional and Deep Learning for EEG-based Epilepsy Detection

2.2.1 Traditional Machine Learning

Before deep learning, the standard pipeline for EEG classification involved extracting hand-crafted features — power spectral density, wavelet coefficients, Hjorth parameters, and so on — and feeding them into a classifier. Shoeb and Guttag [1] demonstrated one of the earliest successful machine learning systems for seizure detection, achieving 96% sensitivity on the CHB-MIT Scalp EEG Database using patient-specific SVM classifiers. Their approach required training a separate model for each patient, which limited scalability but showed that automated detection was feasible. The CHB-MIT dataset they used has since become one of the standard benchmarks in the field, available through PhysioNet [2].

SVMs remained a popular choice for many years due to their effectiveness in high-dimensional feature spaces. The main weakness of this approach is that the quality of results depends heavily on the features chosen, which requires domain expertise and careful tuning. In this project, we include an SVM baseline with wavelet-packet features specifically to quantify how much deep learning adds over this traditional approach.

2.2.2 CNN-based Approaches

CNN-based methods reduced reliance on hand-crafted features by learning directly from raw (or near-raw) EEG data. Acharya et al. [3] proposed a deep 13-layer CNN for seizure detection on the Bonn dataset, achieving 88.67% accuracy on a three-class task (normal, preictal, seizure) with 10-fold cross-validation. Their work was among the first to show that CNNs can learn discriminative EEG patterns without hand-engineered features. However, 88.67% is modest for a three-class problem, and the network’s depth (13 layers) raises questions about whether it was over-parameterised for the small Bonn dataset.

In a related direction, Zhang et al. [4] proposed a cortically subdivided neurodynamic model that uses multi-scale convolutions and physics-informed neural networks (PINNs) for EEG-based Alzheimer’s detection. Although targeting a different condition, their work demonstrates that incorporating domain knowledge about brain structure into the network design can improve neurological signal classification — a principle that influenced our decision to use separate CNN and LSTM components for local and temporal features rather than a single monolithic architecture.

Ullah et al. [5] addressed the parameter-efficiency concern with a pyramidal 1D-CNN (P-1D-CNN), reporting roughly 60% fewer learnable parameters than a conventional CNN. Their study focused particularly on the more difficult ternary Bonn setting and reported accuracies up to 99.1% ($\pm 0.9\%$) across their Bonn experiments after augmentation. The pyramid structure captures multi-scale features through progressively deeper layers. Still, like most CNN-based approaches, temporal dependencies are modelled only implicitly through local convolutions rather than by an explicit sequence module. In our project, the Pure 1D-CNN baseline is used to test whether adding temporal modelling provides additional value beyond local feature extraction alone.

2.2.3 LSTM-based Approaches

Recurrent networks — LSTMs in particular — address the temporal limitation of CNNs. Hussein et al. [6] developed an optimised LSTM architecture specifically designed for robustness against common EEG artefacts such as muscle activity and eye movements. Their key insight was that real clinical EEG is rarely clean, and a model that performs well only on artefact-free data (like the Bonn dataset) may fail in practice. They addressed this by segmenting data and using average pooling before classification. While our project uses the relatively clean Bonn dataset, Hussein’s emphasis on noise robustness is relevant to the future work direction of validating on noisier clinical datasets.

Thara et al. [7] used stacked bidirectional LSTMs on the Bonn dataset, achieving 99.08% detection accuracy (precision 98%, recall 99.5%, AUC 0.984) for binary seizure classification.

Beyond detection, they also attempted seizure *prediction* by classifying preictal states, reaching 89.21% sensitivity — an interesting direction that few other Bonn-based studies explore. Their Bi-LSTM architecture is conceptually close to the recurrent component in our hybrid model.

However, a critical methodological issue exists in their work. Their preprocessing split each 4097-point recording into 23 non-overlapping segments of 178 points, then shuffled all 11,500 segments before doing an 80/20 train-test split. This means segments from the same recording could appear in both training and test sets — a form of data leakage that their paper does not acknowledge. Their proposed wearable wrist-watch device for seizure alarms is a practical vision, but the inflated detection accuracy undermines confidence in the reported performance. Our project explicitly addresses this by enforcing recording-level splitting.

2.2.4 Hybrid Architectures and Recent Work

The idea of combining CNNs with recurrent networks to leverage both local feature extraction and temporal modelling has gained significant traction. Roy et al. [8] proposed ChronoNet, an architecture that combines 1D convolution layers (with exponentially varying filter lengths) and densely connected GRU layers. ChronoNet was evaluated on the Temple University Hospital (TUH) Abnormal EEG Corpus [9] — one of the largest publicly available clinical EEG datasets — and reported state-of-the-art abnormal EEG identification performance at the time. The TUH corpus is substantially larger and noisier than the Bonn dataset, so ChronoNet provides useful evidence that hybrid architectures can remain effective under more realistic data conditions.

Golmohammadi et al. [10] conducted a systematic comparison of LSTM and GRU units within a hybrid CNN-RNN framework, also using the TUH Seizure Corpus. Their finding that convolutional LSTMs significantly outperformed convolutional GRUs supports our choice of LSTM (rather than GRU) as the recurrent component. They also demonstrated that proper initialisation and regularisation are critical for training deep recurrent models on EEG data — a lesson reflected in our use of dropout, early stopping, and learning rate scheduling.

Huang et al. [11] proposed STFFDA, a model combining CNN with 1D-SE attention and Bi-LSTM with dot-product attention, published in *Scientific Reports* in 2025. On the Bonn dataset, they reported 67.24% on five-class classification under 10-fold cross-validation. Their paper also uses a substantially different preprocessing strategy, treating each 4097-point recording as a set of 178-dimensional samples rather than as a continuous time series. For that reason, the comparison with our results is informative but not strictly like-for-like. Even so, the paper remains a useful recent reference because it evaluates five-class performance on the same benchmark with a closely related CNN-BiLSTM-attention family.

Chen et al. [12] took a feature-engineering-heavy approach, combining Discrete Wavelet Transform features with Random Forest selection and CNN classification. They reported 98.47% on binary detection with a 3:1 split, but did not attempt five-class evaluation or cross-validation.

Their use of handcrafted wavelet features before CNN classification is conceptually similar to our SVM baseline, which also uses wavelet-packet features, though with an SVM rather than CNN classifier.

A comprehensive review by Shoeibi et al. [13] surveyed deep learning methods for seizure detection and found that hybrid architectures combining multiple network types tend to outperform single-architecture models, particularly on harder classification tasks. They also noted that most published studies lack systematic ablation experiments, making it unclear which component actually drives performance. This finding directly motivates both the hybrid approach and the ablation study design in this project.

Table 1 summarises the key related works discussed in this section.

Table 1: Summary of Related Work on EEG-based Epilepsy Detection

Study	Method	Year	Evaluation	Task	Accuracy
Shoeb & Gutttag [1]	Patient-specific SVM	2010	Per-patient CV	Binary	96%
Acharya et al. [3]	13-layer CNN	2018	10-Fold CV	3-class	88.67%
Ullah et al. [5]	Pyramidal 1D-CNN	2018	10-Fold CV + aug.	Binary / 3-class	Up to 99.1%
Hussein et al. [6]	Optimised LSTM	2019	Cross-val	Binary	Robust
Thara et al. [7]	Stacked Bi-LSTM	2019	80/20 split	Binary	99.08%
Roy et al. [8]	ChronoNet (CNN+GRU)	2019	TUH dataset	Abnormal	Best at time
Chen et al. [12]	DWT + RF + CNN	2023	3:1 split	Binary	98.47%
Huang et al. [11]	STFFDA	2025	10-Fold CV	5-class	67.24%

2.3 Attention Mechanisms in Biomedical Signal Processing

Not all parts of an EEG recording are equally informative. A 23-second recording might contain a brief 2-second seizure discharge that is critical for classification, while the rest of the signal is unremarkable. Standard pooling operations (global average pooling, for instance) treat every time step the same, diluting the signal from those critical moments.

Self-Attention addresses this by learning a weight for each time step. Given a sequence of Bi-

LSTM outputs $\{h_1, \dots, h_T\}$, the attention mechanism used in this project computes:

$$\begin{aligned}\alpha_t &= \frac{\exp(\mathbf{u}^\top \tanh(\mathbf{W}h_t + b))}{\sum_{j=1}^T \exp(\mathbf{u}^\top \tanh(\mathbf{W}h_j + b))} \\ c &= \sum_{t=1}^T \alpha_t h_t\end{aligned}\tag{1}$$

where $\mathbf{W}$ and $\mathbf{u}$ are learnable parameters and α_t is the attention weight for time step t . The context vector c aggregates the entire sequence into a single representation, weighted by learned importance. This mechanism has been applied successfully in other biomedical domains including ECG classification and EMG analysis.

Huang et al. [11] also used attention in their STFFDA model, but employed a dot-product attention mechanism rather than the additive (Bahdanau-style) attention used here. Their five-class result (67.24%) shows that adding attention does not by itself guarantee strong performance — the quality of the feature representation and the sample construction still matter.

It is worth noting that multi-head self-attention, popularised by the Transformer architecture in natural language processing, has been adapted to EEG in several recent studies. These approaches replace recurrence entirely and model all pairwise time-step interactions in parallel. The trade-off is a quadratic memory cost in the sequence length, which becomes limiting for long EEG windows. In our project, the recurrence-plus-additive-attention combination was preferred because it keeps the memory cost linear in the sequence length while still providing interpretable per-time-step weights for post-hoc analysis (Section 4.3).

2.4 Benchmark Datasets for EEG-based Seizure Detection

The choice of benchmark dataset directly determines which research questions a study can answer. Rather than defaulting to the “largest” or “most clinical” dataset available, a well-designed study should pick the benchmark that isolates the methodological question of interest. For this project, that question is: *does a 1D-CNN + Bi-LSTM + Self-Attention hybrid architecture, under a strict recording-level protocol and against carefully matched baselines, outperform its individual components on epilepsy-related EEG classification?* The Bonn dataset is deliberately chosen as the benchmark most suited to answering this question, for four reasons.

(1) Ablation fairness: controlled, homogeneous recordings. Bonn [14] contains 500 single-channel recordings acquired under tightly controlled conditions (same sampling rate 173.61 Hz, same 23.6 s duration, same electrode configuration per subset, visually artefact-free). This homogeneity is a methodological asset: any accuracy gap between the hybrid model and its pure-CNN / pure-BiLSTM / SVM baselines can be attributed to the architecture itself, not to pre-processing artefacts, channel selection, or inconsistent recording protocols. On noisier datasets, such ablation results are far harder to interpret because preprocessing choices often dominate

the reported differences.

(2) Reproducibility and comparability with prior work. Bonn is the most densely reported benchmark in the EEG-epilepsy literature — Acharya et al. [3], Ullah et al. [5], Thara et al. [7], Chen et al. [12], and Huang et al. [11] all report Bonn numbers for binary, three-class, or five-class tasks. Using the same benchmark makes the comparisons in Section 4.5 directly meaningful and allows this work to identify a specific methodological failure (segment-level leakage) that has inflated accuracy figures across several published studies.

(3) Task structure: a graded difficulty spectrum in a single dataset. Bonn uniquely supports three classification tasks over the same underlying recordings: binary (Z+O vs. S), three-class (Normal vs. Inter-ictal vs. S), and five-class (Z vs. O vs. N vs. F vs. S). This lets the project probe model behaviour along a clear difficulty gradient, from an easy ceiling-effect task to a hard physiologically fine-grained task (N vs. F). No other widely-used single dataset offers this structure out of the box.

(4) Feasibility under a single-student project budget. All model training in this project was run on a single-GPU workstation (NVIDIA RTX 3070 Ti) within a final-year-project time-frame. Under these constraints, a four-model ablation $\times$ three classification tasks $\times$ 10-fold cross-validation is tractable end-to-end on Bonn and leaves room for supplementary experiments (attention visualisation, statistical significance testing). Running the same protocol on a substantially larger corpus such as the TUH Corpus [9] would not only require significantly more compute, but also a non-trivial engineering effort on dataset-specific preprocessing (montage harmonisation, artefact rejection, multi-channel alignment) before any modelling could begin. Choosing Bonn is therefore not a compromise forced by hardware limits, but a design decision that keeps the computational budget spent on the research question itself rather than on infrastructure.

Two other datasets are often cited in the epilepsy EEG literature but are less suited to the methodological focus of this project:

- **CHB-MIT Scalp EEG Database** [1, 2]: long-duration paediatric scalp EEG organised by patient (chb01–chb24). Its native patient-level structure is genuinely useful for subject-wise generalisation studies, but it is also highly heterogeneous and severely imbalanced (seizure events are rare relative to background EEG). This heterogeneity conflates two different research questions — “does the architecture generalise across subjects?” and “does the architecture handle noisy, imbalanced data?” — which makes it a poor choice for the kind of clean ablation study pursued here. CHB-MIT is therefore listed as the natural next step for subject-wise validation in Section 5.3, rather than as the primary benchmark.
- **Temple University Hospital (TUH) EEG Corpus** [9]: the largest publicly available clin-

ical EEG corpus, used by Golmohammadi et al. [10] and Roy et al. [8]. Its scale is useful for large-pretraining approaches, but the preprocessing pipeline required to get consistent results on TUH (montage harmonisation, artefact rejection, heterogeneous channel handling) is itself a significant research effort, one that would compete for time with the core architectural question of this project. Under the single-student timeline available here, TUH-scale experiments would have forced trade-offs on the rigour of the ablation and statistical analysis.

In short, Bonn is not used because larger or more clinical datasets were unavailable; it is used because it is the benchmark that most cleanly isolates the architectural comparison at the heart of this project. CHB-MIT and TUH become more relevant *after* the proposed architecture has been validated on Bonn — which is why they appear in Section 5.3 (Further Work) rather than in the main experiments.

2.5 Research Gaps and Study Novelty

Three gaps in the existing literature motivate this study:

Gap 1: Data leakage in evaluation. Many studies — including Thara et al. [7] — use segment-level or sample-level splitting, where overlapping windows from the same recording can appear in both training and validation sets. This inflates accuracy and makes results hard to trust. Few papers explicitly enforce recording-level splitting. Even recent work like Chen et al. [12] uses simple train-test splits without discussing potential leakage.

Gap 2: No systematic ablation. Most papers compare their model against unrelated baselines. For example, Ullah et al. [5] compare their P-1D-CNN against entirely different architectures from other papers, making it unclear whether improvements come from the model itself or from differences in preprocessing, data splits, or hyperparameters. The review by Shoeibi et al. [13] explicitly noted this as a limitation of the field. A controlled comparison where the same training data and evaluation protocol are used across a hybrid model and its individual components (CNN-only, LSTM-only) is rarely done.

Gap 3: Under-explored multi-class classification. Binary seizure detection is the focus of most work, but it is also the easiest task under the common Bonn Z+O vs. S setup — the signal differences between normal and seizure EEG are large enough that even a traditional SVM achieves 99.67% in this project. The five-class task, which requires distinguishing between different brain states and recording locations, is where model design actually matters. Among the studies reviewed in this chapter, Huang et al. reported 67.24% on Bonn five-class classification under 10-fold cross-validation [11], which still leaves clear room for improvement.

This project addresses all three gaps: strict recording-level cross-validation, a four-model ablation study under identical conditions, and comprehensive evaluation across binary, three-class,

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and five-class tasks with statistical significance testing.

Chapter 3: Design and Implementation

This chapter describes the dataset, preprocessing pipeline, proposed model architecture, baseline models, and experimental setup used in this study.

3.1 University of Bonn EEG Dataset Description

The University of Bonn EEG dataset [14], available through PhysioNet [2], is one of the most widely used benchmark datasets for epileptic seizure detection research. It consists of five subsets (denoted A–E, or equivalently Z, O, N, F, S), each containing 100 single-channel EEG segments of 23.6 seconds duration. All signals were sampled at 173.61 Hz with 12-bit analog-to-digital conversion, resulting in 4097 data points per recording. A hardware bandpass filter with settings of 0.53–40 Hz (12 dB/oct) was applied during acquisition, eliminating the need for additional frequency-domain filtering in preprocessing.

Table 2 summarises the five subsets and their clinical characteristics.

Table 2: University of Bonn EEG Dataset Description

Subset	Label	Description	Recording Type	Count
Z (Set A)	Normal	Healthy volunteers, eyes open	Surface EEG	100
O (Set B)	Normal	Healthy volunteers, eyes closed	Surface EEG	100
N (Set C)	Inter-ictal	Epilepsy patients, seizure-free (contralateral hemisphere)	Intracranial EEG	100
F (Set D)	Inter-ictal	Epilepsy patients, seizure-free (epileptogenic zone)	Intracranial EEG	100
S (Set E)	Ictal	Epilepsy patients, during seizure	Intracranial EEG	100

Three classification tasks of increasing difficulty are defined to comprehensively evaluate the model:

- **Binary classification:** Normal (Z+O, 200 recordings) vs. Seizure (S, 100 recordings).
- **Three-class classification:** Normal (Z+O) vs. Inter-ictal (N+F) vs. Seizure (S).
- **Five-class classification:** Z vs. O vs. N vs. F vs. S (the most challenging task, requiring discrimination between physiologically similar subsets).

Figure 1 illustrates representative EEG waveforms from each subset. Normal recordings (Z, O) exhibit low-amplitude signals typically within $\pm 200 \mu\text{V}$, while seizure recordings (S) display

high-amplitude, rhythmic discharges frequently exceeding $\pm 1000 \mu\text{V}$. The inter-ictal recordings (N, F) show intermediate characteristics with occasional epileptiform discharges.

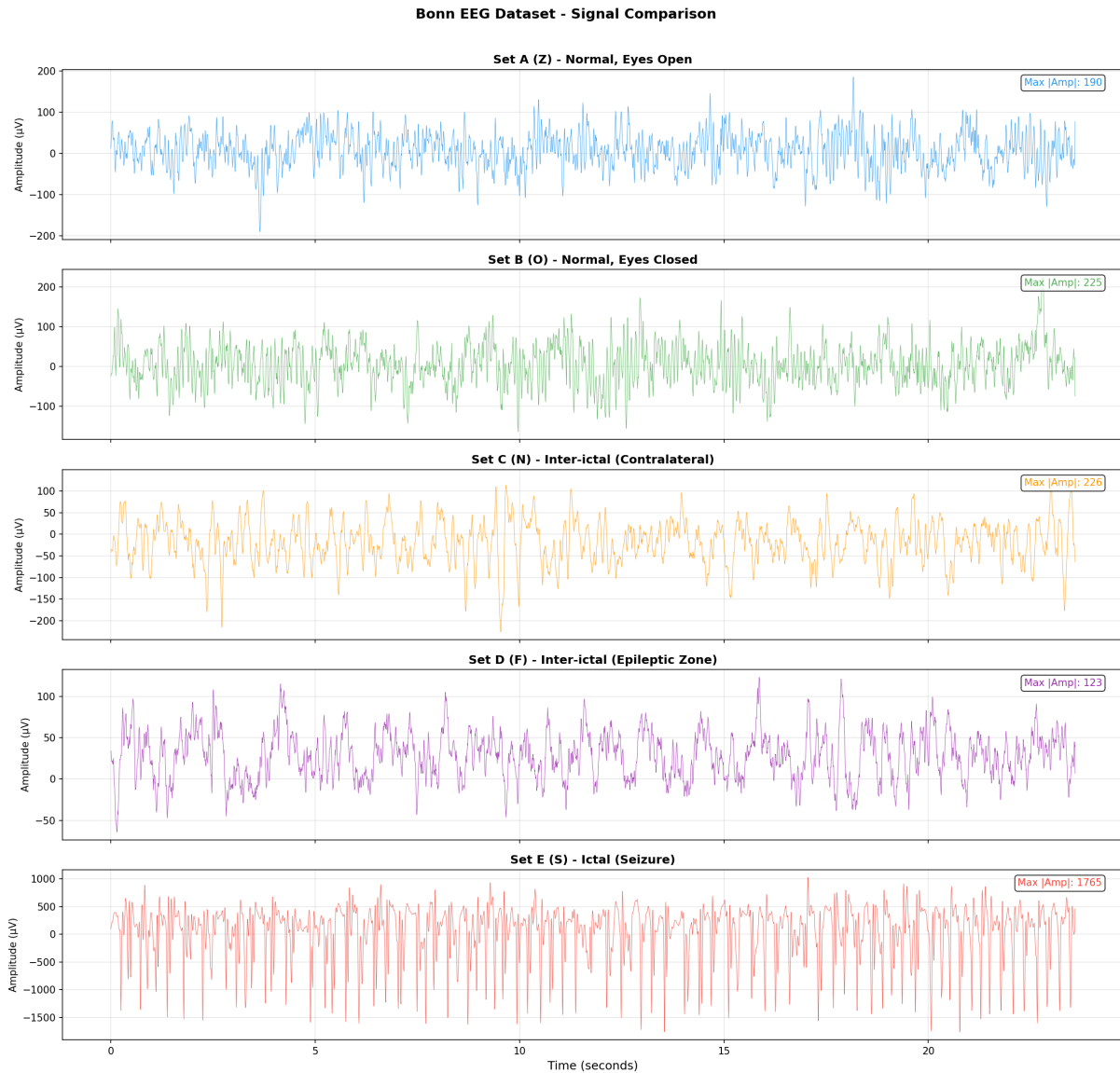


Figure 1: Representative EEG signal waveforms from the five subsets of the Bonn dataset. Seizure (S) signals show distinctly higher amplitudes and rhythmic patterns compared to normal (Z, O) signals.

The frequency-domain characteristics further highlight the differences between subsets. Figure 2 shows the Fast Fourier Transform (FFT) magnitude spectra for each class. Seizure recordings concentrate energy in the low-frequency delta and theta bands (0.5–8 Hz), while normal recordings show broader spectral distributions with prominent alpha-band (8–13 Hz) activity, particularly in the eyes-closed condition (Set O).

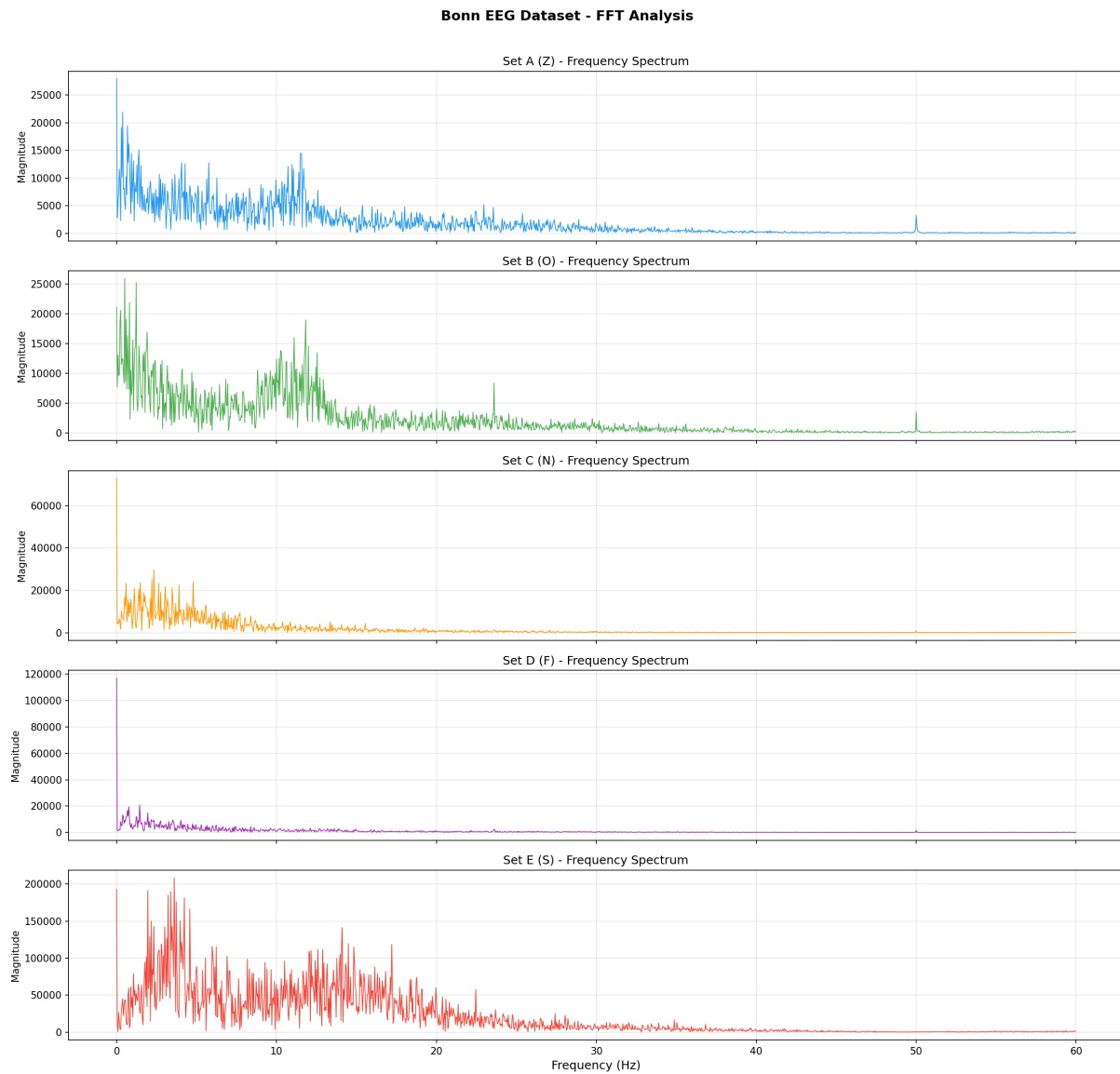


Figure 2: FFT frequency spectra for the five EEG subsets, revealing distinct spectral signatures

Figure 3 shows the amplitude distributions for each subset. The clear separation between Normal ($<200 \mu V$) and Seizure ($>1000 \mu V$) amplitudes explains why binary classification is straightforward, while the significant overlap between N and F distributions explains why five-class classification is much harder.

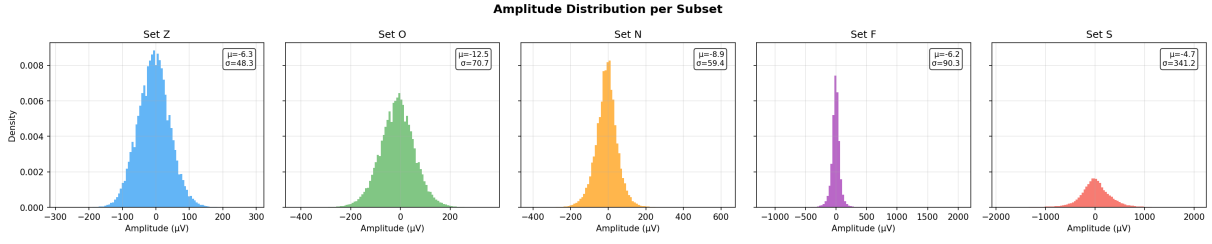


Figure 3: Amplitude distribution histograms for the five EEG subsets

3.2 EEG Signal Preprocessing Pipeline

The preprocessing pipeline was designed with particular emphasis on preventing data leakage, a methodological concern raised by the project supervisor and commonly overlooked in related literature.

3.2.1 Recording-level K-Fold Cross-Validation

A strict recording-level 10-Fold Stratified Cross-Validation strategy is employed. In each fold, the dataset is first split at the *recording* level — all sliding-window segments derived from a single original EEG recording are kept entirely within either the training or validation set, never split across both. This prevents the model from “seeing” overlapping segments of the same recording during both training and evaluation.

This approach differs from the segment-level splitting used by many prior studies (e.g., Thara et al. [7]), where individual segments are shuffled independently, allowing fragments of the same recording to appear in both training and validation sets. Such leakage inflates reported accuracy, particularly when segments overlap.

3.2.2 Data Leakage Discovery and Correction

An earlier version of this project’s pipeline applied sliding-window segmentation *before* the train/validation split, which constituted a data leakage vulnerability. Combined with a globally-fitted StandardScaler, this resulted in artificially inflated accuracy (99.95% on binary classification). After identifying this issue, the pipeline was refactored to ensure:

1. Recording-level splitting is performed *first*.
2. Sliding-window segmentation is applied *independently* to training and validation recordings.

3. The `StandardScaler` is fitted exclusively on training data, then applied to transform validation data.

After correction, the binary accuracy decreased from 99.95% to 99.67%, and the five-class accuracy from 86.60% to 84.51%, which is consistent with the removal of optimistic bias introduced by the earlier leakage-prone pipeline.

3.2.3 Sliding Window Segmentation

Each 4097-point recording is segmented using a sliding window of 1024 data points (≈ 5.9 seconds at 173.61 Hz) with 50% overlap (stride = 512 points). This produces approximately 7 segments per recording and expands the effective dataset from 500 original recordings to approximately 3,500 training samples, mitigating the risk of overfitting on the small raw dataset.

3.2.4 Z-Score Normalisation

A global Z-Score normalisation is applied using `StandardScaler`: a single mean and standard deviation are computed across all data points in the training set of each fold, then used to transform both training and validation data. This approach preserves inter-window amplitude differences — an important discriminative feature, as seizure signals exhibit significantly higher amplitudes than normal recordings.

Figure 4 summarises the complete preprocessing and evaluation pipeline.

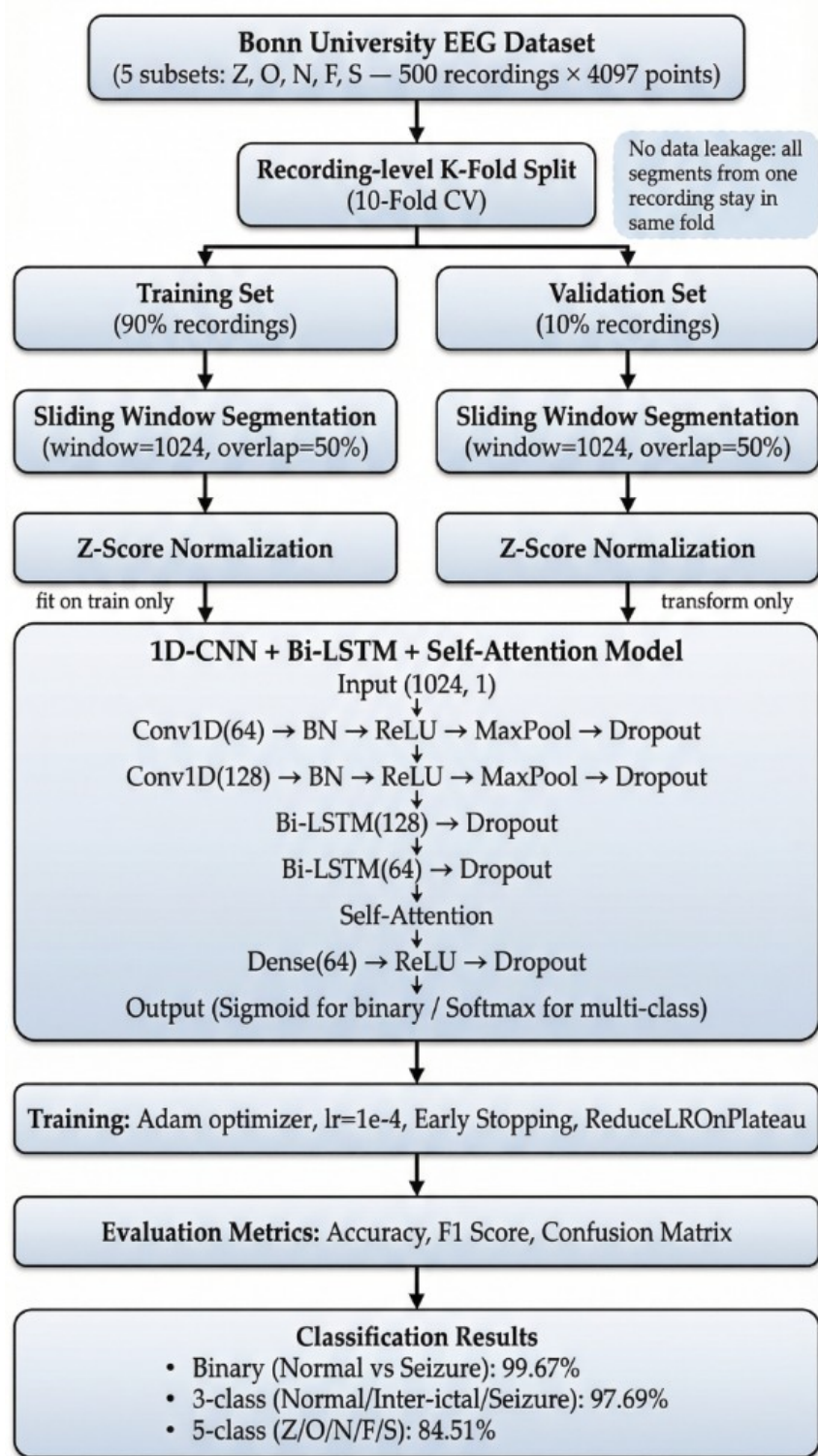


Figure 4: Preprocessing and evaluation pipeline. The red dashed box highlights the recording-level split that prevents data leakage. Sliding window segmentation and normalisation are applied independently to training and validation sets.

3.3 Proposed Hybrid Model Architecture

The proposed model is a hybrid deep learning architecture combining three complementary components: a 1D Convolutional Neural Network (1D-CNN) for local feature extraction, a Bidirectional Long Short-Term Memory network (Bi-LSTM) for temporal dependency modelling, and a Self-Attention mechanism for focusing on discriminative time steps. Table 3 summarises the architecture.

Table 3: Proposed Hybrid Model Architecture

Component	Configuration	Role
1D-CNN Layer 1	Conv1D(1→64, kernel=7), BN, ReLU, MaxPool(2), Dropout(0.2)	Local feature extraction
1D-CNN Layer 2	Conv1D(64→128, kernel=7), BN, ReLU, MaxPool(2), Dropout(0.2)	Higher-level local patterns
Bi-LSTM Layer 1	128 hidden units, bidirectional, Dropout(0.4)	Forward/backward temporal modelling
Bi-LSTM Layer 2	64 hidden units, bidirectional, Dropout(0.4)	Refined temporal representation
Self-Attention	64-dim, tanh scoring, softmax weighting	Focus on key time steps
Classifier	Dense(128→64), ReLU, Dropout(0.4), Output	Classification head

Figure 5 provides a visual overview of the hybrid model architecture, showing the data flow from input through each component to the final classification output.

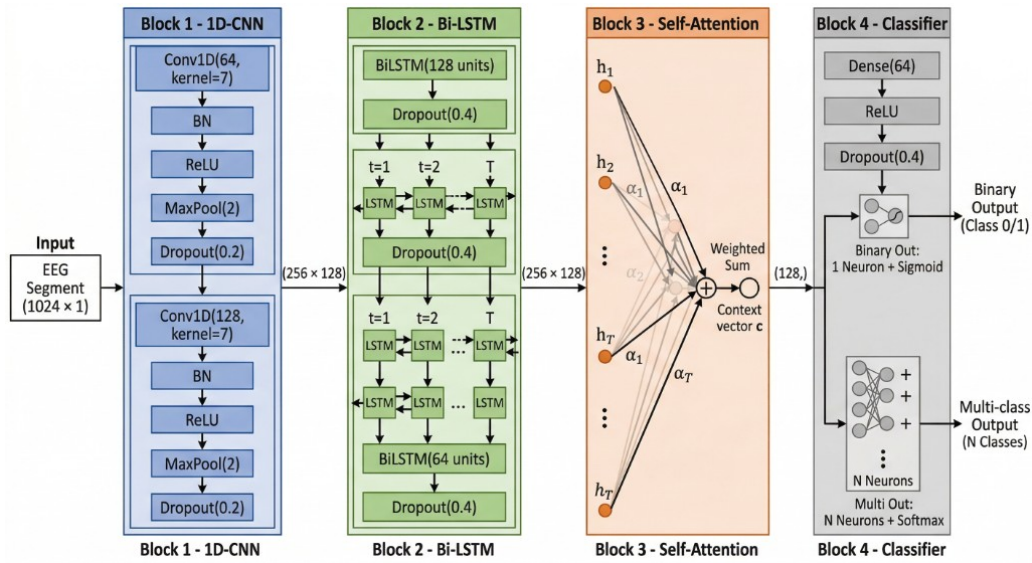


Figure 5: Architecture of the proposed hybrid 1D-CNN + Bi-LSTM + Self-Attention model. Tensor shapes are annotated at each stage. The CNN extracts local features (blue), the Bi-LSTM models temporal dependencies (green), the Self-Attention assigns importance weights (orange), and the classifier produces the final output (grey).

3.3.1 1D-CNN Feature Extraction

The 1D-CNN module consists of two convolutional layers with 64 and 128 filters respectively, each with a kernel size of 7. Each layer is followed by Batch Normalisation, ReLU activation, max-pooling (factor 2), and dropout (rate 0.2). The CNN extracts local spatio-temporal features such as spike-wave complexes and sharp transients that are characteristic of epileptic activity. The two max-pooling operations reduce the temporal resolution by a factor of 4, compressing the 1024-point input to a 256-step feature sequence.

3.3.2 Bidirectional LSTM

The output of the CNN is passed to two stacked Bi-LSTM layers with 128 and 64 hidden units respectively. Each Bi-LSTM processes the sequence in both forward and backward directions, producing output vectors of dimension $2 \times \text{hidden_units}$ at each time step. This bidirectional processing captures both past and future temporal context, which can help model the evolution of seizure dynamics over time.

3.3.3 Self-Attention Mechanism

The Self-Attention layer computes an importance weight α_t for each time step t of the Bi-LSTM output h_t :

$$e_t = \tanh(\mathbf{W}h_t + b) \quad (2)$$

$$\alpha_t = \frac{\exp(\mathbf{u}^\top e_t)}{\sum_{j=1}^T \exp(\mathbf{u}^\top e_j)} \quad (3)$$

$$c = \sum_{t=1}^T \alpha_t h_t \quad (4)$$

where $\mathbf{W} \in \mathbb{R}^{128 \times 64}$ and $\mathbf{u} \in \mathbb{R}^{64}$ are learnable parameters. The context vector c is a weighted sum of all time steps, allowing the model to focus on the most discriminative temporal regions rather than treating all time steps equally.

3.3.4 Output Layer

For binary classification, the classifier outputs a single logit, which is optimised with Binary Cross-Entropy with Logits Loss; the sigmoid transformation is therefore handled inside the loss function rather than by a separate output activation. For three-class and five-class tasks, the output layer has N neurons (where N is the number of classes) and is trained with Cross-Entropy Loss, which applies softmax implicitly. The total number of trainable parameters is approximately 504,000 (1.92 MB).

3.4 Baseline Models for Comparative Analysis

Three baseline models were implemented to serve as an ablation study, systematically evaluating the contribution of each component in the hybrid architecture. Table 4 summarises the baselines and their roles.

Table 4: Baseline Models for Ablation Study

Baseline	Architecture	Ablation Purpose
SVM + Wavelet Packet	RBF-SVM on 64-dim WPD features (Daubechies-4, 4 levels)	Traditional ML reference
Pure 1D-CNN	4-layer CNN (32→64→128→128) + Global Average Pooling	Remove temporal modelling
Pure Bi-LSTM + Attention	2-layer Bi-LSTM (128, 64) + Self-Attention (no CNN)	Remove local feature extraction (no CNN)

The SVM baseline uses Wavelet Packet Decomposition to extract 16 sub-band nodes, from which four statistical features (energy, mean, standard deviation, and Shannon entropy) are computed per node, yielding a 64-dimensional feature vector per recording. A Radial Basis Function (RBF) kernel SVM is then trained on these features. Notably, the SVM operates on complete recordings without sliding-window segmentation, making its K-Fold split inherently at the recording level.

3.5 Experimental Setup and Training Strategy

Table 5 lists the training hyperparameters used for all deep learning models.

Table 5: Training Hyperparameters

Parameter	Value
Optimiser	Adam
Learning rate	1×10^{-4}
LR scheduler	ReduceLROnPlateau (factor=0.5, patience=7)
Early stopping patience	15 epochs
Maximum epochs	100
Batch size	64
Cross-validation	10-Fold Stratified, recording-level
Random seed	42

All deep learning experiments were conducted on an NVIDIA GeForce RTX 3070 Ti GPU (8 GB VRAM) using PyTorch 2.x with CUDA acceleration. The framework was migrated from TensorFlow/Keras because native Windows GPU support was discontinued in TensorFlow 2.11 and later versions. Under the local setup used in this project, that change enabled GPU training and reduced the measured wall-clock time from roughly ~ 55 s to ~ 0.7 s per training epoch.

The evaluation metrics used throughout this study are:

- **Accuracy:** The proportion of correctly classified samples.
- **Weighted F1 Score:** The harmonic mean of precision and recall, weighted by class support, to account for potential class imbalance.
- **Confusion Matrix:** Visualisation of per-class classification performance.
- **Paired t-test:** Statistical significance testing across cross-validation folds to determine whether observed performance differences are statistically reliable ($p < 0.05$).

Chapter 4: Results and Discussion

This chapter presents the experimental results of the proposed hybrid model and three baseline models across all classification tasks, followed by error analysis, attention visualisation, statistical significance testing, and a comparison with published literature.

All results in this chapter come from strict recording-level 10-Fold cross-validation with no data leakage. Metrics are reported as mean $\pm$ standard deviation across the 10 folds.

4.1 Classification Performance across All Tasks

4.1.1 Binary Classification: Normal (Z+O) vs. Seizure (S)

Table 6 presents the binary classification results. All four models achieve accuracy above 98.8%, reflecting the substantial signal-level differences between normal EEG ($<200 \mu\text{V}$ amplitude) and seizure activity ($>1000 \mu\text{V}$). This ceiling effect is consistent with findings reported in the literature: Thara et al. [7] reported 99.08% under an 80/20 segment-level split, while Ullah et al. [5] reported up to 99.1% using a pyramidal CNN.

Table 6: Binary Classification Results — Normal (Z+O) vs. Seizure (S)

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	99.67 (± 1.00)	99.66 (± 1.01)
Pure 1D-CNN	98.86 (± 1.91)	98.84 (± 1.95)
Pure Bi-LSTM + Attention	99.48 (± 1.29)	99.47 (± 1.29)
CNN + Bi-LSTM + Attention (Ours)	99.67 (± 0.85)	99.66 (± 0.86)

The hybrid model ties with SVM at 99.67% accuracy but achieves the lowest reported variance ($\pm 0.85\%$ vs. $\pm 1.00\%$), indicating the most consistent cross-fold performance in the current runs.

Figure 6 shows the combined 10-fold training curves. The model converges rapidly — most folds reach near-perfect validation accuracy within 20 epochs, with early stopping terminating training shortly after. The tight clustering of fold curves reflects the consistency of this task.

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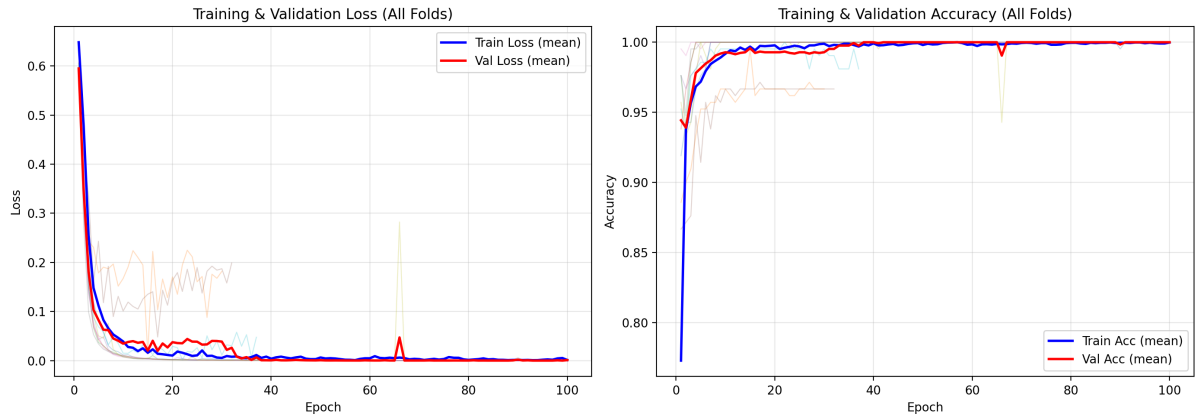
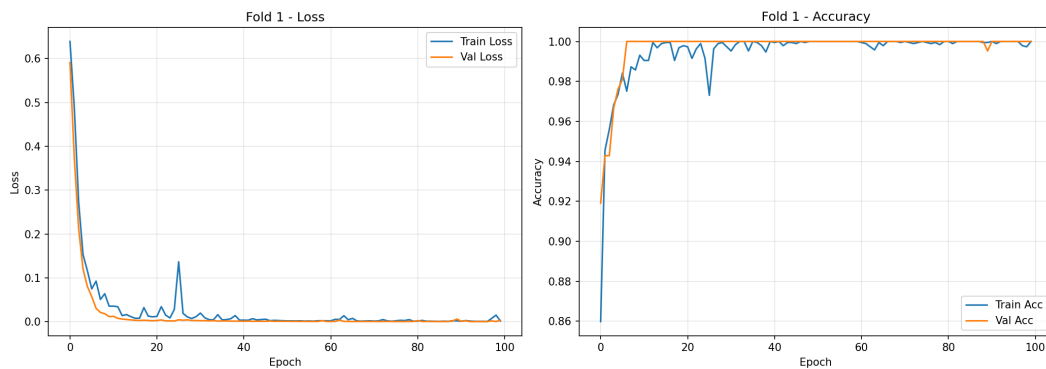


Figure 6: Combined 10-fold training curves for binary classification. Light lines: individual folds; bold line: mean.

Figure 7 compares the best and worst folds. Fold 1 converges to 100% within 15 epochs. Fold 6 is the lowest-performing fold (97.1%), where validation loss remains slightly elevated.

(a) Fold 1 — best performance (100%)



(b) Fold 6 — lowest performance (97.1%)

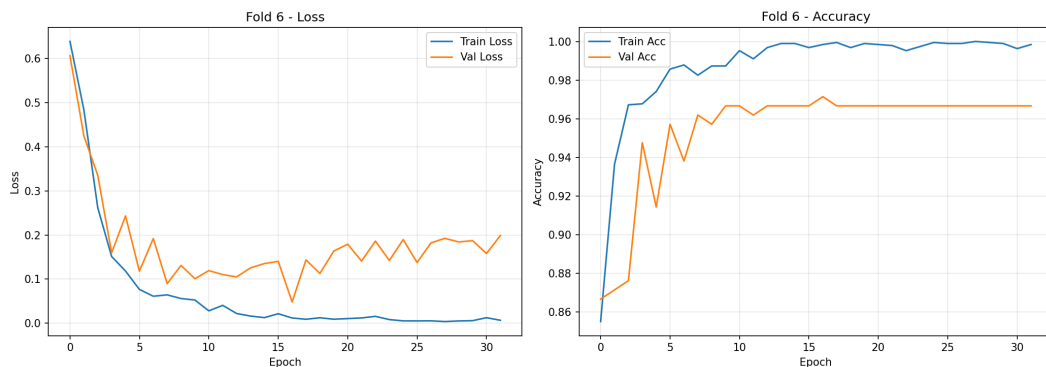


Figure 7: Individual fold training curves for binary classification, showing the best and worst folds.

4.1.2 Three-class Classification: Normal vs. Inter-ictal vs. Seizure

Table 7 shows the three-class results. The introduction of the inter-ictal class increases task difficulty, as inter-ictal EEG shares some morphological features with both normal and seizure recordings.

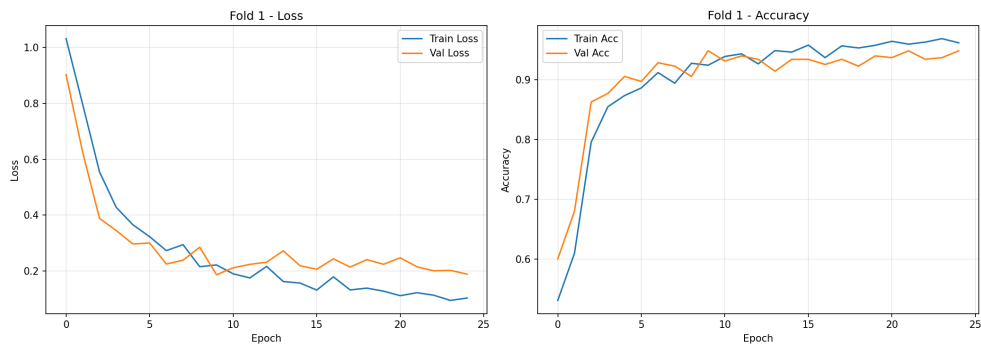
Table 7: Three-class Classification Results — Normal vs. Inter-ictal vs. Seizure

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	94.20 (± 2.44)	94.19 (± 2.44)
Pure 1D-CNN	97.14 (± 1.84)	97.11 (± 1.86)
Pure Bi-LSTM + Attention	92.80 (± 5.99)	92.74 (± 6.12)
CNN + Bi-LSTM + Attention (Ours)	97.69 (± 2.17)	97.68 (± 2.17)

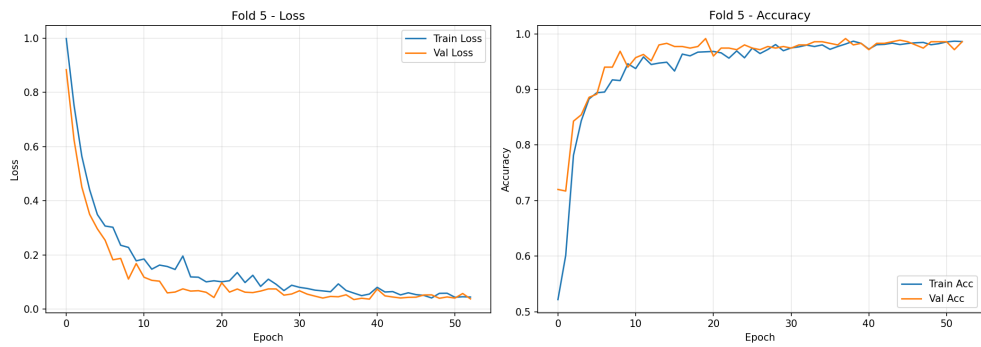
The hybrid model achieves the highest accuracy at 97.69%, outperforming SVM by +3.49% and Pure Bi-LSTM by +4.89%. The Pure Bi-LSTM exhibits notably high variance ($\pm 5.99\%$), suggesting that raw-signal sequence modelling without CNN-based local feature preprocessing is less stable in this setting. The Pure CNN performs competitively at 97.14%, only 0.55% below the hybrid model.

Combined 10-fold curves were not generated for the three-class task during the original experiments. Figure 8 shows representative folds. Folds 1 and 5 converge above 97% validation accuracy, while Folds 9 and 10 plateau at 94–95%, pulling the overall mean to 97.69%.

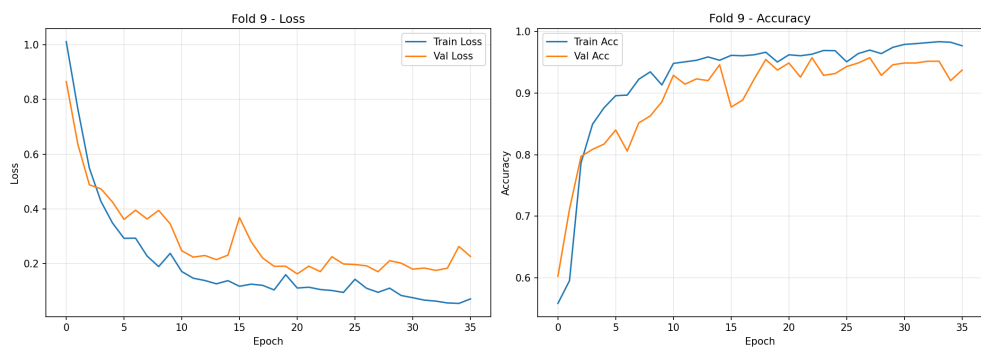
(a) Fold 1 — converges above 97%



(b) Fold 5 — converges above 97%



(c) Fold 9 — plateaus around 94%



(d) Fold 10 — plateaus around 95%

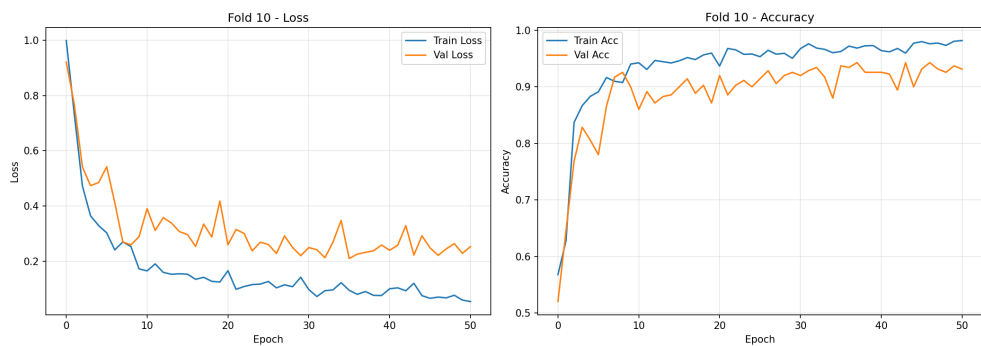


Figure 8: Three-class training curves — representative individual folds. (a)(b) high-performing folds converging above 97%; (c)(d) lower-performing folds plateauing around 94–95%.

4.1.3 Five-class Classification: Z vs. O vs. N vs. F vs. S

Table 8 presents the most challenging task. The five-class task requires the model to distinguish between physiologically similar subsets, particularly between N and F (both inter-ictal recordings from different brain regions) and between Z and O (both normal recordings under different eye conditions).

Table 8: Five-class Classification Results — Z vs. O vs. N vs. F vs. S

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	78.80 (± 5.31)	78.39 (± 5.69)
Pure 1D-CNN	78.86 (± 3.77)	75.95 (± 5.12)
Pure Bi-LSTM + Attention	73.09 (± 8.37)	71.92 (± 9.24)
CNN + Bi-LSTM + Attention (Ours)	84.51 (± 3.62)	84.28 (± 3.80)

The hybrid model achieves 84.51% accuracy, outperforming Pure CNN by +5.65%, SVM by +5.71%, and Pure Bi-LSTM by +11.42%. The hybrid model also exhibits the lowest standard deviation ($\pm 3.62\%$) among the deep learning models, indicating the lowest cross-fold variance in this task.

Figure 9 shows the combined 10-fold training curves. Convergence takes longer than binary or three-class (typically 40–60 epochs), and the spread between folds is noticeably wider, consistent with the $\pm 3.62\%$ standard deviation in Table 8.

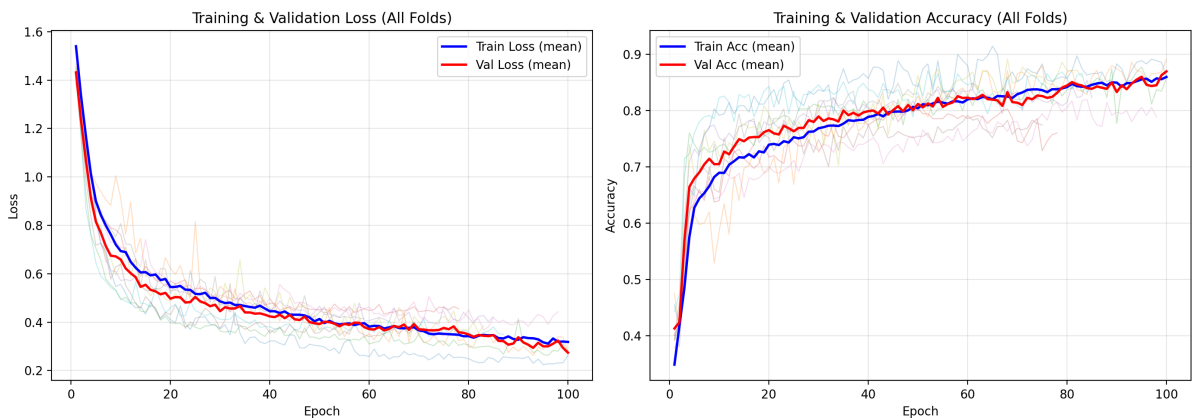


Figure 9: Combined 10-fold training curves for five-class classification. The wider spread between folds reflects the greater difficulty of this task.

Figure 10 compares Fold 1 (88.9%, one of the best) with Fold 4 (78.9%, one of the worst). The 10% gap illustrates how recording-level data partitions can affect five-class performance, particularly the N/F separation.

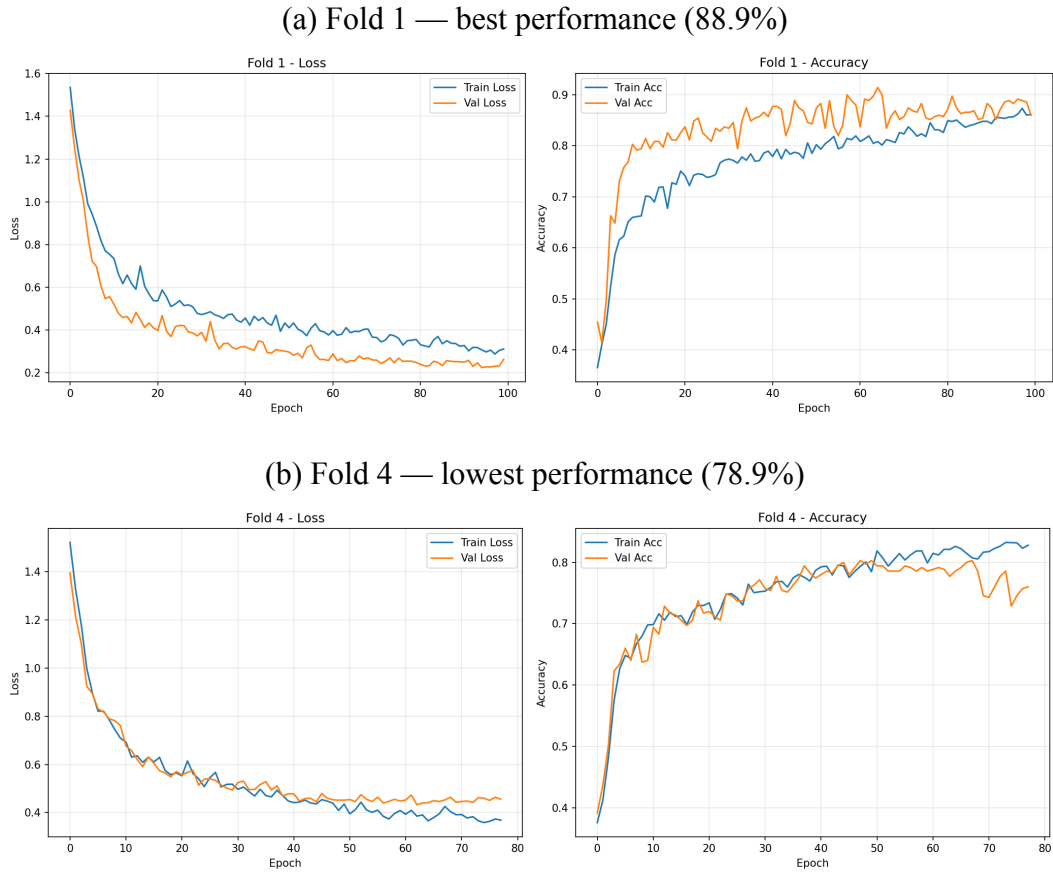


Figure 10: Individual fold training curves for five-class classification, showing the best and worst folds.

4.2 Confusion Matrix and Error Pattern Analysis

Figure 11 shows the aggregated confusion matrix for the hybrid model on binary classification. The result is near-perfect: out of all test samples across 10 folds, only a handful of seizure samples were misclassified as normal. This confirms the ceiling effect discussed in Section 4.1.1.

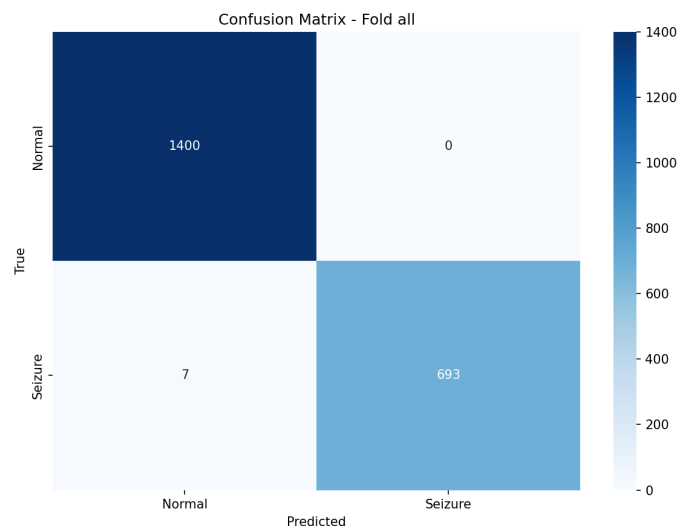


Figure 11: Aggregated confusion matrix for the hybrid model on binary classification (Normal vs. Seizure)

Figure 12 shows the three-class result. The model separates all three classes well, though a small number of inter-ictal samples are confused with the normal class.

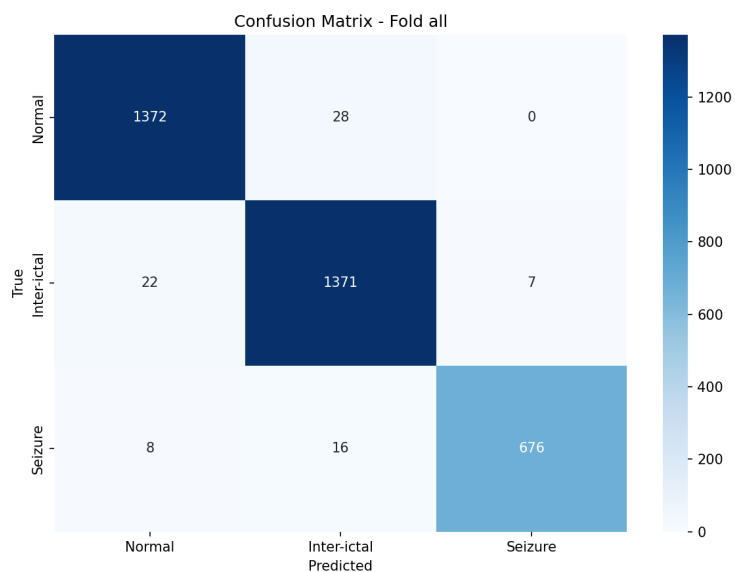


Figure 12: Aggregated confusion matrix for the hybrid model on three-class classification

The five-class task (Figure 13) reveals the primary error source: **confusion between Set N and Set F**. Both subsets contain inter-ictal EEG recordings from epilepsy patients — Set N from the hippocampal formation contralateral to the epileptogenic zone, and Set F from within the epileptogenic zone itself. Their physiological similarity makes them difficult to distinguish and is likely a major reason why five-class performance remains well below binary performance. A secondary source of error is mild confusion between Set Z (eyes open) and Set O (eyes closed),

which differ mainly in the presence or absence of the alpha rhythm. In contrast, Set S (seizure) is almost never misclassified.

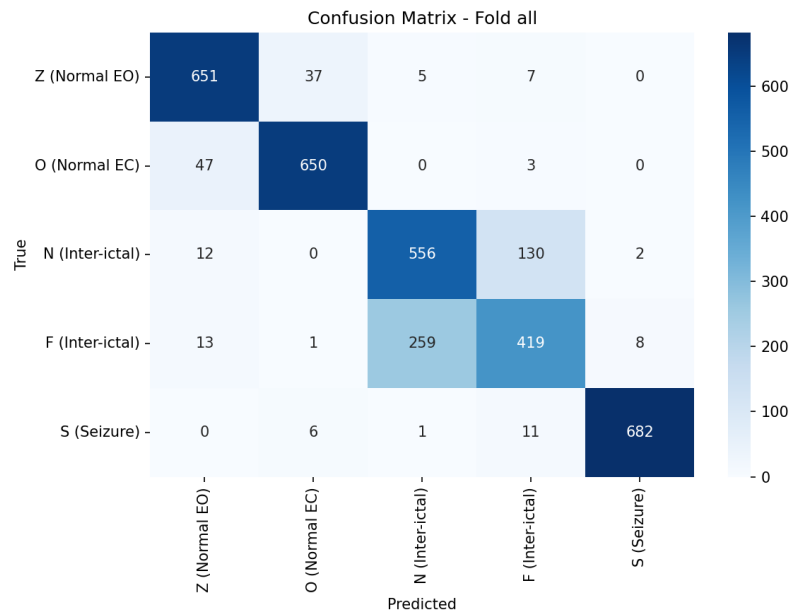


Figure 13: Aggregated confusion matrix for the hybrid model on five-class classification. The $N \leftrightarrow F$ confusion is the dominant error source.

To understand how the baselines differ, Figure 14 compares the five-class confusion matrices of all four models. The SVM and Pure CNN show similar overall patterns to the hybrid model but with more errors spread across the off-diagonal cells. The Pure Bi-LSTM shows notably worse separation — its N and F columns are heavily confused, and even the Z/O distinction degrades. The hybrid model achieves the cleanest diagonal, particularly for the N and F classes where CNN-based local feature extraction appears to be important.

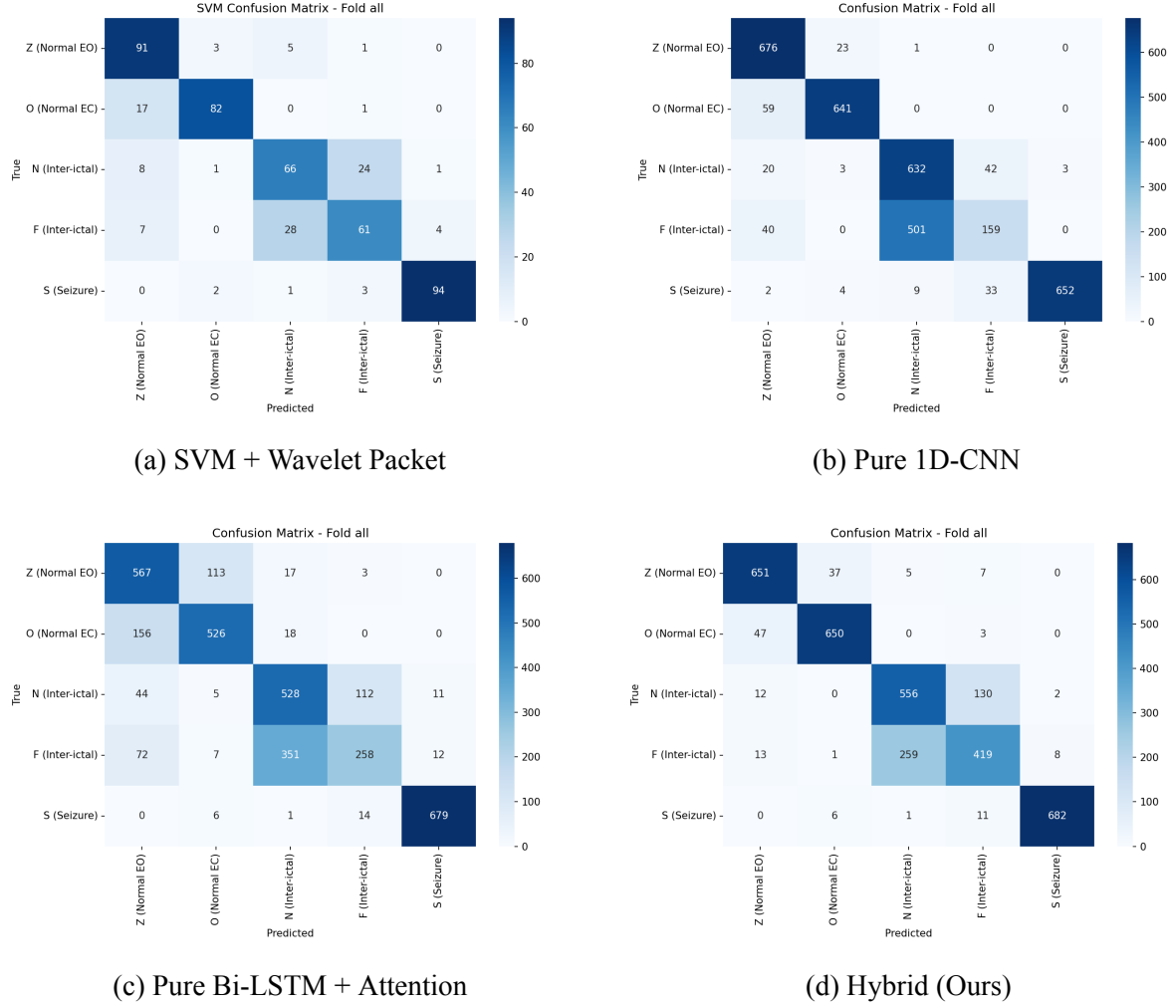


Figure 14: Comparison of five-class confusion matrices across all four models. The hybrid model (d) shows the cleanest diagonal with fewest off-diagonal errors.

4.3 Attention Weight Visualisation and Interpretability

To investigate the interpretability of the proposed model, the learned Self-Attention weights were extracted and visualised for several representative EEG recordings. Figure 15 shows the attention weight distribution overlaid on the normalised EEG signal for four representative recordings.

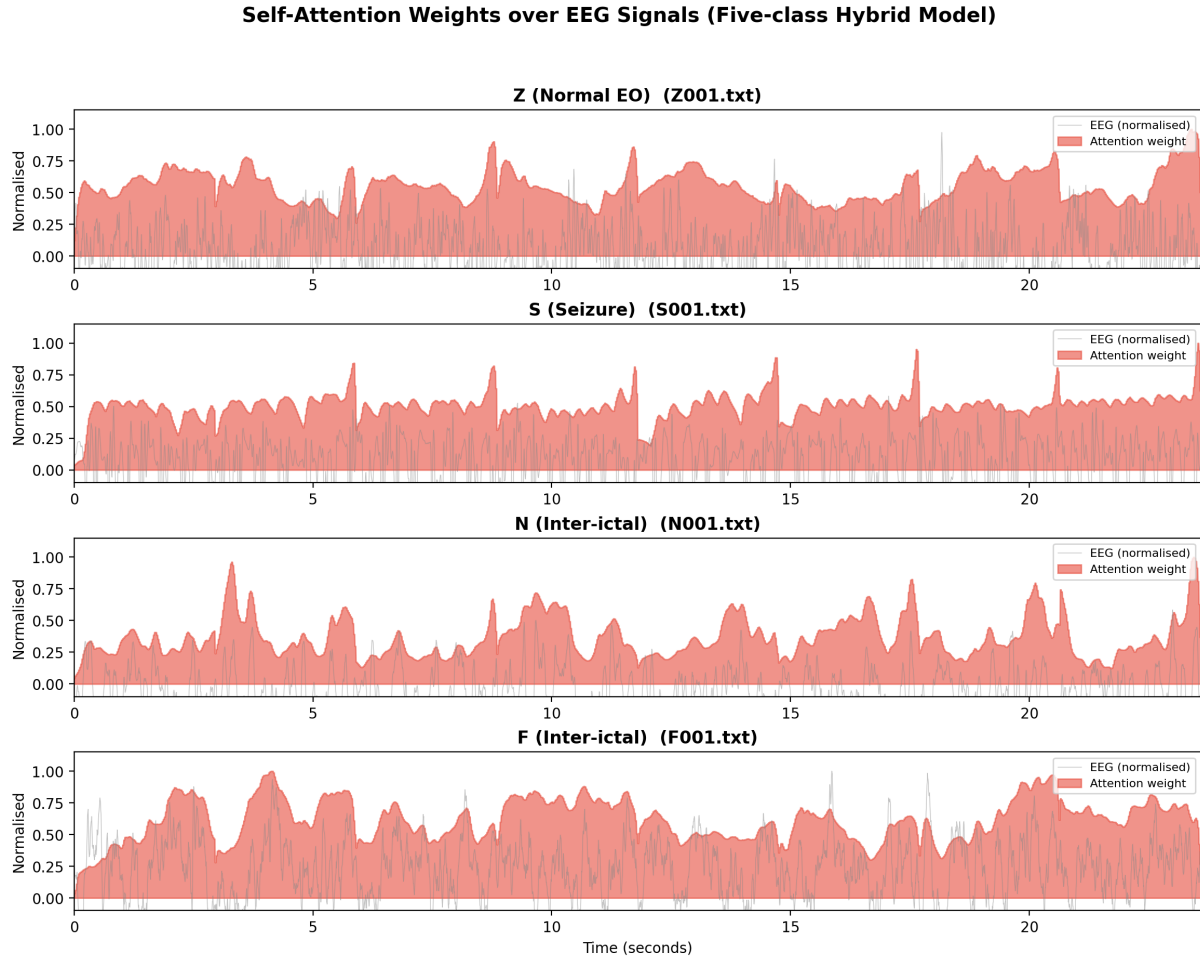


Figure 15: Self-Attention weight visualisation for representative EEG recordings from the five-class hybrid model. The normalised EEG signal is shown in grey, with attention weights overlaid in red. Higher red intensity indicates greater model focus on that temporal region.

The visualisation reveals distinct attention patterns across different EEG classes:

- **Seizure recordings (S):** Attention is concentrated on the high-amplitude, rhythmic discharge regions, which are the hallmark of ictal activity. This aligns with the clinical criteria used by neurologists for seizure identification.
- **Normal recordings (Z):** Attention is distributed more uniformly across the signal, reflecting the absence of pathological transients.
- **Inter-ictal recordings (N, F):** Attention shows intermediate patterns, with occasional peaks corresponding to interictal epileptiform discharges (IEDs).

These patterns are consistent with the idea that the Self-Attention mechanism assigns higher weights to salient temporal regions. However, the visualisation should be interpreted as qualitative evidence about model behaviour rather than as a clinically validated explanation.

4.4 Statistical Significance Testing

To determine whether the observed performance differences are statistically reliable, paired t-tests were conducted across the 10 cross-validation folds for each pair of models within each task. Table 9 presents the key results.

Table 9: Paired t-test Results (Selected Comparisons)

Task	Comparison	t -statistic	p -value	Sig.?
Binary	Hybrid vs. SVM	0.000	1.000	No
Binary	Hybrid vs. CNN	-1.403	0.194	No
Three	Hybrid vs. SVM	4.327	0.002	Yes
Three	Hybrid vs. BiLSTM	-2.324	0.045	Yes
Three	CNN vs. SVM	3.950	0.003	Yes
Five	Hybrid vs. CNN	-5.247	0.001	Yes
Five	Hybrid vs. BiLSTM	-5.289	0.001	Yes
Five	Hybrid vs. SVM	2.205	0.055	No

The statistical analysis reveals three key findings:

1. **Binary task:** Among the reported pairwise tests for the binary task, no significant differences were observed ($p > 0.05$), which is consistent with the ceiling effect discussed in Section 4.1.1. In the Bonn Z+O vs. S setting, several reasonable approaches reach near-perfect performance.
2. **Five-class task:** The hybrid model *significantly* outperforms both the Pure CNN ($p = 0.001$) and Pure Bi-LSTM ($p = 0.001$), supporting the view that combining CNN and Bi-LSTM components improves performance beyond random variation in this setting.
3. **Hybrid vs. SVM on five-class:** Despite a 5.71% mean accuracy advantage, the difference does not reach statistical significance ($p = 0.055$). This is attributable in part to the high inter-fold variance of SVM ($\pm 5.31\%$), where SVM outperformed the hybrid model on 2 out of 10 folds. A larger study would be needed before claiming a significant advantage over SVM.

4.5 Comparison with Published Literature

Table 10 compares the proposed model with recently published methods on the Bonn EEG dataset.

Table 10: Comparison with Published Literature on the Bonn Dataset

Method	Year	Task	Evaluation	Acc.
Thara et al. [7] (Stacked Bi-LSTM)	2019	Binary	Single split (80/20)	99.08%
Ullah et al. [5] (Pyramidal CNN)	2018	Binary / 3-class	10-Fold CV + aug.	Up to 99.1%
Chen et al. [12] (RF + CNN)	2023	Binary	3:1 split	98.47%
Huang et al. [11] (STFFDA)	2025	5-class	10-Fold CV	67.24%
Ours (CNN+BiLSTM+Attn)	2026	Binary	10-Fold recording-level	99.67%
Ours (CNN+BiLSTM+Attn)	2026	3-class	10-Fold recording-level	97.69%
Ours (CNN+BiLSTM+Attn)	2026	5-class	10-Fold recording-level	84.51%

The proposed model performs strongly on the evaluated Bonn tasks. On the five-class task, this work reports 84.51%, which is higher than the 67.24% reported by Huang et al. [11]. The comparison should still be read with care, because the preprocessing and sample construction differ between the two studies. Even so, the result suggests that the current hybrid pipeline remains competitive under a stricter recording-level split.

Three methodological choices probably contribute to this gap. First, the hybrid model keeps each 1024-point window as a continuous time series rather than flattening it into fixed-length feature vectors, which preserves the temporal context that the Bi-LSTM and attention components rely on. Second, recording-level splitting — while it eliminates overlap-induced data leakage — does not prevent the model from learning useful generalisable features; the model still reaches 84.51% without any obvious leakage signal, which suggests that performance is driven by genuine pattern recognition rather than by memorising specific recordings. Third, the ablation protocol in Section 4.1 shows that no single component (CNN-only, Bi-LSTM-only) reaches the full hybrid accuracy, which is consistent with the interpretation that both local spectral features and longer-range temporal structure are needed for the five-class decision boundary.

4.6 Discussion of Failure Cases and Per-class Observations

Beyond the aggregated accuracy numbers, several per-class observations from the confusion matrices in Section 4.2 deserve an explicit discussion, because they reveal where the hybrid model is most and least reliable.

Seizure (S) class: near-perfect detection. Across all three classification tasks and all four

models, Set S is almost never misclassified. This is not surprising given the amplitude gap between ictal and non-ictal EEG (roughly an order of magnitude on the Bonn dataset), but it is a useful sanity check: if the model failed to detect the most visually obvious class, it would signal a pipeline bug rather than a modelling shortcoming.

Inter-ictal N vs. F: the dominant five-class confusion. Set N (inter-ictal, contralateral to the epileptogenic zone) and Set F (inter-ictal, within the epileptogenic zone) are both recorded from epilepsy patients in their seizure-free state. The neurophysiological difference between these two classes is subtle: it reflects the spatial origin of the recording electrodes relative to the seizure focus, rather than any global difference in the signal. The fact that the hybrid model still achieves usable N/F separation is encouraging, but the residual confusion in the $N \leftrightarrow F$ cells of Figure 13 is the single largest contributor to the drop from 97.69% (three-class) to 84.51% (five-class). A multi-channel dataset with explicit spatial information would likely narrow this gap.

Normal Z vs. O: a smaller but consistent confusion. Set Z (eyes open) and Set O (eyes closed) differ mainly in the presence or absence of the posterior alpha rhythm (8–13 Hz). Because the Bonn recordings are single-channel, this alpha difference is the only cue available, and the Bi-LSTM + attention component does pick it up in most folds. The few misclassifications here are plausibly caused by recordings where the alpha rhythm is less pronounced, e.g., under mild drowsiness.

Fold-to-fold variance. The individual fold curves in Figures 7, 8, and 10 show that not all folds are equally easy. For binary classification, Fold 6 (97.1%) is a clear outlier; for five-class, Fold 4 (78.9%) is the lower bound. Inspection of the fold assignments suggests that folds with a higher concentration of F recordings in the validation set tend to be the more difficult ones, which is consistent with the N/F confusion pattern above. This cross-fold variance is a natural consequence of recording-level splitting on a small dataset, and it is a useful reminder not to over-interpret any single-fold number.

4.7 Clinical Applicability and Limitations

The proposed system is better viewed as a proof-of-concept for offline EEG classification than as a deployable clinical decision support tool. The strong binary result on Bonn (99.67%) suggests that the model can separate clearly different signal classes under controlled conditions, but whether this would transfer to routine EEG monitoring remains to be tested.

The attention visualisation in Section 4.3 offers a limited interpretability signal by highlighting which temporal regions received higher weights, but this should not be treated as direct clinical validation.

However, several limitations must be acknowledged:

- **Single dataset:** All experiments were conducted on the Bonn dataset, which is relatively small (500 recordings) and clean (pre-filtered, artefact-free). Performance on noisy, real-world clinical recordings may differ.
- **Single-channel:** The Bonn dataset contains only single-channel EEG, whereas modern clinical EEG typically uses multi-channel montages (19+ channels). The model’s ability to leverage spatial information from multi-channel data remains unexplored.
- **Absence of per-file subject identifiers:** The Bonn dataset was recorded from 5 subjects per subset, but the public release does not provide a per-file mapping from recordings to subject identities [14]. As a result, all cross-validation in this work is performed at the recording level rather than at the subject level. A strict subject-wise (leave-subjects-out) evaluation — which would better reflect the case of deploying the model on previously unseen patients — is therefore not feasible with Bonn alone. This limitation was raised during supervision, and the agreed arrangement is to retain the recording-level protocol for the Bonn-based evaluation and to plan a subject-wise evaluation on a dataset with explicit subject IDs as future work (see Section 5.3).
- **Class imbalance:** The binary task involves a 2:1 class ratio (200 Normal vs. 100 Seizure recordings). No explicit class weighting was applied, and no clear degradation was observed in the reported binary results, but the imbalance should still be considered when generalising to more imbalanced clinical datasets.
- **Offline analysis:** The current system processes pre-recorded EEG segments and is not optimised for real-time streaming inference.

Chapter 5: Conclusion and Further Work

5.1 Conclusion

This project set out to build a hybrid deep learning model for automated epilepsy detection from EEG signals and to evaluate it under a stricter protocol than the initial implementation. The final results support the usefulness of that design on the Bonn benchmark.

The proposed 1D-CNN + Bi-LSTM + Self-Attention architecture achieves 99.67% on binary classification, 97.69% on three-class, and 84.51% on the five-class task — all evaluated under strict recording-level 10-Fold cross-validation with no data leakage. On Bonn five-class classification, the reported mean accuracy is higher than the 67.24% reported by Huang et al. [11], although the preprocessing pipelines differ and the comparison should therefore be treated as indicative rather than exact.

The ablation study suggests that the three components contribute complementary strengths. Paired t-tests show the hybrid model significantly outperforms both the pure CNN ($p = 0.001$) and pure Bi-LSTM ($p = 0.001$) on five-class classification. The binary task, by contrast, shows no significant differences in the reported pairwise tests, which is consistent with a ceiling effect in this particular Bonn setup.

Attention weight visualisation highlights high-amplitude seizure regions and offers a qualitative view of model behaviour. It is useful for discussion, but it should not be treated as a clinically validated explanation.

5.2 Reflection

Looking back, the most useful lesson from this project concerned evaluation methodology more than model design.

Early in the project, my initial preprocessing pipeline produced near-perfect accuracy — over 99.9% on binary classification. The numbers looked appealing, but my supervisor rightly questioned them. After investigation, I found that overlapping sliding-window segments from the same recording were leaking across the train/validation split. Fixing this dropped accuracy by about 0.3% on binary and 2% on five-class. The revised numbers are less dramatic, but they are more credible. This part of the project taught me that evaluation design matters at least as much as model design.

On the technical side, migrating the codebase from TensorFlow to PyTorch mid-project was not planned but turned out to be necessary — TensorFlow 2.11 dropped native Windows GPU support, and under my local setup training on CPU was far slower than PyTorch with CUDA. The migration forced me to understand both frameworks at a deeper level than I would have

otherwise.

I also want to be transparent about the role of AI tools. I used Cursor as a support tool for code scaffolding, debugging, and some report drafting. All generated code and text were reviewed and revised by me. The main project decisions — including the model choice, the leakage correction, and the experimental design — remained my responsibility.

5.3 Further Work

Several directions could extend this work:

- **Subject-wise (patient-independent) evaluation on a dataset with explicit subject IDs.** A limitation of the current evaluation, as discussed in Section 4.6, is that the Bonn release does not include a per-file subject-ID mapping, so all cross-validation in this work is at the recording level rather than the subject level. After consulting the supervisor, the agreed plan is to address this as future work using CHB-MIT [1], which is organised by patient (e.g., chb01–chb24) and natively supports a leave-subjects-out protocol (for example, 6 patients for training and 4 for testing). This would test whether the reported accuracy holds when the model is evaluated on previously unseen patients — a setting closer to real clinical deployment.
- **Larger, multi-channel datasets.** The Bonn dataset has only 500 single-channel recordings and is relatively clean. Validating on noisier, multi-channel datasets like CHB-MIT [1] or the Temple University Hospital EEG Corpus [9] would test whether the model generalises to real clinical conditions.
- **Hyperparameter sensitivity analysis.** The current hyperparameters (window size 1024, dropout 0.4, LSTM units 128/64) were chosen based on literature defaults and preliminary experiments. A systematic grid search or Bayesian optimisation over these parameters could improve performance or confirm robustness.
- **Cross-dataset transfer learning.** Pre-training on a large, diverse EEG corpus and fine-tuning on a smaller target dataset could improve generalisation in data-scarce clinical settings.
- **Real-time inference.** The current system processes pre-recorded segments. Deploying it for real-time EEG monitoring would require model compression (pruning, quantisation) and optimisation for edge hardware.
- **Patient-specific adaptation.** EEG patterns vary substantially between individuals. Developing mechanisms for the model to adapt to a specific patient’s baseline could improve detection accuracy in personalised settings.

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Acknowledgment

I would like to express my sincere gratitude to my project supervisor for their patient guidance, constructive feedback, and encouragement throughout this project. In particular, their insistence on methodological rigour — including the identification of data leakage in my initial pre-processing pipeline — was instrumental in shaping the quality of this work.

I would also like to thank the staff of the International School at Beijing University of Posts and Telecommunications and the School of Electronic Engineering and Computer Science at Queen Mary University of London for providing the academic environment and resources that made this project possible.

Finally, I am grateful to my family and friends for their unwavering support during this demanding final year.

Appendices

Disclaimer

This report is submitted as part requirement for the undergraduate degree programme at Queen Mary University of London, and Beijing University of Posts and Telecommunications. It is the product of my own labour except where indicated in the text. The report may be freely copied and distributed provided the source is acknowledged.

BUPT No.: 2022213356

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Full Name (Pin Yin): Zhong Ziwen

Full Name (Chinese): 钟梓文

Signature: 

Date: 27-04-2026

Acknowledgement of using Generative AI

I acknowledge the use of the following Generative AI tool:

Publisher	Name, version, URL	Use	Section
Anysphere	Cursor IDE, 2026, https://www.cursor.com/	To assist with debugging the data leakage issue in the preprocessing pipeline, refactoring code for recording-level cross-validation, and formatting LaTeX tables and figures in the report.	Chapter 3, Chapter 4, code scripts

I confirm the submitted work is my own work.

北京邮电大学 本科毕业设计（论文）任务书

Project Specification Form

Part 2 - Student

学院 School	International School	专业 Programme	Internet of Things Engineering		
姓 Family name	Zhong	名 First Name	Ziwen		
BUPT 学号 BUPT number	2022213356	QM 学号 QM number	221168280	班级 Class	2022215112
论文题目 Project Title	AI Assisted Epilepsy detection using EEG signals				
论文概述 Project outline Write about 500-800 words Please refer to Project Student Handbook section 3.2	Problem Analysis: Epilepsy is a neurological disease affecting almost 50 million people worldwide. Recurrent seizures can cause injury, cognitive decline, and social stigma. The core problem is: how to detect seizures automatically and accurately from scalp EEG in real clinical settings, where signals are noisy, patients vary in age and recording conditions, and traditional manual review is time-consuming and subjective. This project aims to develop a Deep Learning Network (DLN) model to automatically classify EEG signals as epileptic or non-epileptic, thereby assisting neurologists to diagnose faster and more accurate. Why Deep Learning (DL)? Traditional machine learning relies on handcrafted features (e.g., wavelet energy), which are sensitive to noise and require expert tuning. DL, however, learns hierarchical representations directly from raw EEG, automating feature extraction and classification. The 2021 review by Shoeibi et al. confirms that DL consistently outperforms traditional methods on public EEG datasets, especially in multi-channel, long-duration recordings. Data Collection: Data will be sourced from publicly available anonymized EEG datasets (e.g., CHB-MIT Scalp EEG Database or Temple University Hospital EEG Corpus), ensuring no direct interaction with patients. All data is pre-collected, pre-processed, and anonymized, eliminating the need for informed consent. Proposed Solution: The core of my epilepsy detection system is a hybrid deep learning architecture combining 1D Convolutional Neural Networks (1D-CNN), Bidirectional Long Short-Term Memory (Bi-LSTM) layers, and a Self-Attention mechanism. This design is specifically tailored to the unique challenges of multi-channel EEG signals—high-dimensional, non-stationary time series with subtle, transient pathological patterns that traditional methods often fail to capture reliably. My understanding of EEG dynamics and deep learning principles drives this choice: seizures are not isolated events but evolve over time with localized spikes, rhythmic discharges, and channel-specific correlations, requiring a model that extracts hierarchical features while preserving temporal context and focusing on clinically meaningful segments.				

	- 1D-CNN extracts local spatio-temporal patterns (e.g., spike-wave complexes) across channels. - Bi-LSTM models long-term temporal dependencies (seizure evolution over seconds). - Self-Attention dynamically focuses on seizure onset windows, enhancing interpretability. This hybrid architecture is ideal for multi-channel, variable-length EEG and has shown >98% accuracy in recent cross-dataset studies. Implementation Plan: 1.Pre-processing (MNE-Python): - Resample all data to 256 Hz (unifies sampling, preserves <45 Hz seizure bands). - Apply 0.5–45 Hz bandpass filter (removes drift, EMG, power-line noise). - Z-score per channel (normalizes amplitude across patients/devices). - Segment into 4-second windows, 50% overlap (captures full seizure dynamics, increases data density). 2.Training: - Pre-train on 80% of TUH (large, diverse). - Fine-tune on CHB-MIT with domain adaptation - Optimizer: AdamW, lr=1e-4, early stopping. 3.Baselines: - SVM + wavelet-packet features (traditional ML). - Pure 1D-CNN (no LSTM/Attention) — to validate sequential modeling. 4.Evaluation: 10-fold patient-level cross-validation; metrics: Accuracy, Sensitivity, Specificity, F1, AUC. 5.Output: Trained .h5 model, performance tables. Results will be exported as PDF reports. Tools and Hardware: Language: Python 3.9+ Libraries: MNE-Python, TensorFlow with Keras API, Scikit-learn, Matplotlib Hardware: Google Colab Pro (GPU), EECS JupyterHub, local laptop for development Dataset: Public EEG datasets (anonymized) References: 1. Acharya UR, Oh SL, Hagiwara Y, Tan JH, Adeli H. Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals. Comput Biol Med. 2018 Sep 1;100:270-278. 2. Acharya UR, et al. Automated EEG-based screening of depression using deep convolutional neural network. Comput Methods Programs Biomed. 2018;161:103–13. 3. Goldberger AL, et al. PhysioBank, PhysioToolkit, and PhysioNet. *Circulation*. 2000;101(23):e215–20. 4. Obeid I, Picone J. The Temple University Hospital EEG Data Corpus. *Front Neurosci*. 2016;10:196. 5. Shoeb AH, Gutttag JV. Application of machine learning to epileptic seizure detection. In: Proceedings of the 27th International Conference on Machine Learning (ICML'10). 2010. p. 975–982.
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道德规范 Ethics	Please confirm by checking the box: ☒ I confirm that I have discussed ethical issues with my supervisor.

Please discuss ethical issues with your supervisor. Please refer to Project Student Handbook section 4.1 and Project Ethics Guidance	Summary of ethical issues: I have discussed the ethical implications of this project with my supervisor. This project does not involve human participants or the collection of primary data from individuals. All EEG data will be obtained from publicly available, fully anonymized datasets (e.g., CHB-MIT, TUH EEG Corpus), which have been pre-approved for research use. However, as the project uses Artificial Intelligence (AI) techniques, the following ethical considerations apply: Data Provenance: All datasets and code will be properly cited using Vancouver referencing. No data or code will be used without attribution. Data Confidentiality: Datasets are fully anonymised at source. No personally identifiable information will be processed. Data will be stored securely on password-protected cloud storage (Google Drive) and deleted at the end of the project. Addressing Bias: Class imbalance and demographic limitations (e.g., paediatric focus in CHB-MIT) will be analysed and mitigated through stratified sampling, weighted loss function, and ablation studies. Societal Impact: The system is intended solely for research and diagnostic support. It will not be deployed in clinical settings without further validation. Potential misuse (e.g., over-reliance leading to misdiagnosis) will be acknowledged.
中期目标 Mid-term target. It must be tangible outcomes, E.g. software, hardware or simulation. It will be assessed at the mid-term oral.	Mid-term target (March 2025) By the mid-term assessment, the following tangible outcomes will be achieved: 1. Dataset acquired and pre-processed: Full EEG dataset downloaded, filtered, segmented, and split into train/validation/test sets. 2. DLN model implemented: Proposed DLN model built in TensorFlow/Keras with working training loop. 3. Initial training completed: Model trained with recorded loss/accuracy curves. 4. Preliminary results: Accuracy, sensitivity, and confusion matrix reported on validation set.

Work Plan (Gantt Chart)

Fill in the sub-tasks and insert a letter X in the cells to show the extent of each task

	Nov 1-15	Nov 16-30	Dec 1-15	Dec 16-31	Jan 1-15	Jan 16-31	Feb 1-15	Feb 16-28	Mar 1-15	Mar 16-31	Apr 1-15	Apr 16-30
Task 1 [To learn Deep Learning Network and EEG signal classification]												
Literature review on DLN and EEG epilepsy	X	X	X									
Study MNE-Python and EEG preprocessing	X	X	X	X								
Datasets collection and exploration		X	X	X	X							
Task 2 [To develop the DLN model for classifying Epilepsy patient EEG signals]												
Design 1D-CNN architecture (DLN)			X	X	X							
Implement preprocessing pipeline				X	X	X						
Data harmonization and cleaning					X	X	X					
Task 3 [To train the Deep Learning model with the EEG data]												
Implement 5-fold stratified cross-validation					X	X	X					
Train 1D-CNN with merged dataset						X	X	X	X			
Hyperparameter tuning and ablation study							X	X	X	X		
Task 4 [To compare the performance of DLN model with other DL models]												
Implement LSTM baseline								X	X	X		
Implement SVM with spectral features									X	X	X	
Performance comparison and statistical analysis										X	X	X

北京邮电大学 本科毕业设计（论文） 初期进度报告

Project Early-term Progress Report

学院 School	International School	专业 Programme	Internet of Things Engineering		
姓 Family name	Zhong	名 First Name	Ziwen		
BUPT 学号 BUPT number	2022213356	QM 学号 QM number	221168280	班级 Class	2022215112
论文题目 Project Title	AI Assisted Epilepsy detection using EEG signals				
1. Literature Review and Critical Analysis Epilepsy is a chronic non-communicable disorder of the brain that affects approximately 50 million people worldwide. The hallmark of epilepsy is recurrent seizures, which are brief episodes of involuntary movement. EEG remains the gold standard for diagnosis. However, manual inspection of long-term EEG recordings is tedious, time-consuming, and prone to human error. Consequently, Automated Seizure Detection Systems (ASDS) using Deep Learning have gained significant attention. In preparing for this project, I have conducted a systematic review of recent literature, focusing on the evolution from traditional machine learning to advanced Deep Learning architectures. My analysis highlights a progression from feature-engineering-heavy approaches to end-to-end learning models. 1.1 Convolutional Neural Networks (CNN) in EEG Analysis Convolutional Neural Networks have been extensively and successfully applied to EEG signal analysis. Unlike their application in static images, 1D-CNNs in EEG processing utilize convolutional filters to extract local temporal and spectral features effectively. One of the foundational works in this domain is by Acharya et al. (2017), who proposed a deep Convolutional Neural Network for the automated detection of seizure segments[1]. Their model utilized a 13-layer deep CNN architecture to automatically learn features from raw EEG signals, achieving high accuracy. Recent Developments: Advanced architectures such as CodimNet[2] demonstrate the power of convolutional filtering in capturing complex non-linear features from raw EEG data without manual feature engineering. CodimNet illustrates how multi-scale convolutions can handle the high-dimensional nature of EEG signals robustly. My Perspective: While CNN-based approaches like CodimNet are highly effective at feature extraction, they primarily focus on local patterns within the receptive fields. For this specific project, I aim to investigate Recurrent Neural Networks, specifically Bi-LSTM, to explore their capability in modeling long-term temporal dependencies across the entire seizure event. This provides a comparative perspective to the dominant CNN-based methods. 1.2 The Shift to Recurrent Neural Networks (RNN) and LSTM Addressing the temporal limitations of CNNs, Hussein et al. (2018) introduced a robust Long Short-Term Memory (LSTM) network [3]. Their approach focused on analyzing EEG signals that contain artifacts (muscle activities, eye movements) and white noise. They employed a strategy of segmenting the dataset and using average pooling before the final classification to handle the non-stationary nature of the signals. My Insight & Critique: The most valuable takeaway from Hussein's research is the importance of robustness against noise. In clinical settings, EEG data is rarely clean. However, standard LSTMs					

process data in a single direction (past to future). In my view, for a retrospective analysis of EEG recordings (where the entire signal is available), limiting the model to uni-directional context is inefficient. We should exploit the context from both past and future data points to make a more informed classification for any given timestamp.

1.3 Bidirectional and Hybrid Architectures

Building on the concept of temporal context, Thara et al. (2019) proposed a Stacked Bidirectional LSTM (Bi-LSTM) model [4]. Their study on the Bonn dataset achieved 99% accuracy by stacking two layers of Bi-LSTMs, allowing the network to learn from the input sequence in both forward and backward directions.

My Insight & Critique: This paper is pivotal for my project. The success of Bi-LSTM confirms my hypothesis that bidirectional information flow is critical for high-fidelity seizure detection. However, a gap remains: deep models like stacked Bi-LSTMs are computationally expensive. Ullah et al. (2018) proposed a Pyramidal 1D-CNN to reduce parameter size (using only 39% of traditional parameters) while maintaining accuracy [5].

Synthesis for My Project: Balancing accuracy and computational efficiency is key. Therefore, my project will focus on optimizing a Bi-LSTM architecture specifically for the Bonn dataset. Unlike many studies that merge multiple datasets (e.g., CHB-MIT with Bonn) to increase data volume, I will strictly adhere to the project constraint of using the Bonn University Dataset alone. This forces the challenge of designing a model that generalizes well on a smaller, specific dataset without overfitting—a problem I intend to solve by implementing rigorous data augmentation and regularization techniques (like Dropout), as suggested in the literature.

2. Summary of Progress

I have successfully completed the initial phase of the project, moving from theoretical understanding to practical environment setup and data analysis.

2.1 Theoretical Foundation and Skill Acquisition

Deep Learning Architectures: I have completed an in-depth study of Recurrent Neural Networks and their variants. Specifically, I focused on the "Vanishing Gradient Problem" inherent in simple RNNs and how the Gating mechanisms in LSTMs (Input, Forget, and Output gates) and GRUs (Update and Reset gates) solve this.

EEG Signal Characteristics: I familiarized myself with the physiological basis of EEG, understanding the difference between *ictal* (seizure), *inter-ictal* (between seizures), and *normal* states, and how these manifest in signal amplitude and frequency.

2.2 Dataset Acquisition and Exploratory Data Analysis (EDA)

I have selected the **University of Bonn EEG Dataset** as the data source for this project.

- **Data Structure:** I successfully acquired the complete dataset, which consists of 5 subsets (Z, O, N, F, S), also referred to as Sets A, B, C, D, and E.
 - **Sets A & B:** Surface EEG from healthy volunteers (Eyes Open/Closed).
 - **Sets C & D:** Intracranial EEG from epilepsy patients during seizure-free intervals (Inter-ictal).
 - **Set E:** Intracranial EEG during seizure activity (Ictal).
- **Data Processing:** The original data was in single-channel ASCII format. I wrote a Python script using pandas and numpy to:
 1. Iterate through the 100 text files in each folder.
 2. Merge them into a structured dataset of size 500×4097 samples.
 3. Visualize the signals using matplotlib.
- **Key Observation:** My initial visualization confirms distinct differences in amplitude. Set E (Seizure) shows significantly higher amplitudes (often $> 1000 \mu V$) and chaotic

patterns compared to Sets A and B ($< 100 \mu V$). This validates that the dataset is suitable for training a discriminative Deep Learning model.

2.3 Technical Implementation

Environment Setup: I have configured a local development environment using Anaconda. The core stack includes Python 3.8, TensorFlow 2.x (Keras API) for model building, and Scikit-learn for data splitting and evaluation metrics.

Preprocessing Pipeline: I have begun implementing a segmentation function. Since each recording is 23.6 seconds (4097 data points), feeding the entire sequence into an LSTM might cause memory bottlenecks. I am currently experimenting with segmenting these into smaller windows (e.g., 2048 or 1024 data points) to increase the number of training samples and improve the model's ability to detect short-duration anomalies.

3. Strategy for Addressing the Small Dataset Issue

A critical challenge in this project is the limited size of the Bonn University Dataset (500 trials total). Training deep learning models on such small data can lead to severe overfitting. To address this, I have formulated a comprehensive **Data Augmentation and Regularization Strategy**:

3.1 Data Segmentation with Overlap (Sliding Window)

Instead of feeding the entire 23.6-second recording (4097 data points) as a single sample, I will implement a **Sliding Window Segmentation** technique.

- **Implementation:** The continuous EEG signal will be sliced into smaller windows (e.g., 1 to 2 seconds).
- **Overlap:** Crucially, I will apply a **50% overlap** between consecutive windows.
- **Impact:** This technique will increase the dataset size from 500 samples to approximately **10,000+ training segments**. This massive increase in effective sample size is essential for the model to learn generalized features and converge properly.

3.2 K-Fold Cross-Validation

To ensure the reported accuracy is reliable and not a result of a lucky train-test split, I will employ **10-Fold Cross-Validation**. The dataset will be divided into 10 subsets; the model will be trained on 9 and tested on 1, repeating the process 10 times. The final performance will be the average of these 10 runs, providing a statistically robust metric.

3.3 Regularization Techniques

- **Dropout:** I will insert Dropout layers (rate = 0.3 to 0.5) within the Bi-LSTM architecture to randomly drop neurons during training, forcing the network to learn redundant representations and preventing reliance on specific noise patterns.
- **Early Stopping:** Training will monitor the validation loss and stop automatically when performance plateaus, preventing the model from memorizing the training data.

4. References

- [1] U. R. Acharya, S. L. Oh, Y. Hagiwara, J. H. Tan, and H. Adeli, "Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals," *Computers in Biology and Medicine*, vol. 100, pp. 270–278, Sep. 2017.
- [2] Z. Zhang, Y. Wei, X. Rao, *et al.*, "Electroencephalographic biomarker-guided early detection of Alzheimer's disease via cortically subdivided neurodynamic PINN," *Expert Syst. Appl.*, vol. 297, p. 129353, 2026.
- [3] R. Hussein, H. Palangi, R. K. Ward, and Z. J. Wang, "Optimized deep neural network architecture for robust detection of epileptic seizures using EEG signals," *Clinical Neurophysiology*, vol. 130, no. 1, pp. 25–37, 2019.
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- [5] I. Ullah, M. Hussain, E. Qazi, and H. Aboalsamh, "An automated system for epilepsy detection using EEG brain signals based on deep learning approach," *Expert Systems with Applications*, vol. 107, pp. 61–71, 2018.
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- [7] M. Golmohammadi et al., "Gated Recurrent Networks for Seizure Detection," *Neural Engineering Data Consortium*, Temple University, Philadelphia, PA, USA, 2019.

是否符合进度? On schedule as per GANTT chart?

YES. I have completed the literature review, selected the methodology (Bi-LSTM), acquired and preprocessed the Bonn dataset, and set up the coding environment. This aligns perfectly with the initial phase of the project timeline.

下一步 Next steps:

1. Construct and compile the Bi-Directional LSTM model using TensorFlow/Keras.
2. Perform initial training on the 2-class problem (Normal vs. Seizure) and record baseline metrics.
3. Implement data segmentation techniques to handle the 4097-point sequences effectively.

北京邮电大学 本科毕业设计（论文）中期进度报告

Project Mid-term Progress Report

学院 School	International School	专业 Programme	Internet of Things Engineering																				
姓 Family name	Zhong	名 First Name	Ziwen																				
BUPT 学号 BUPT number	2022213356	QM 学号 QM number	221168280	班级 Class	2022215112																		
论文题目 Project Title	AI Assisted Epilepsy detection using EEG signals																						
是否完成任务书中所定的中期目标? Targets met (as set in the Specification)? YES, I have completed the mid-term targets specified in the project specification.																							
已完成工作 Finished work: All the four predefined mid-term targets in the project specification have been fully accomplished, and the project progress has even exceeded the original plan, with additional baseline model implementation and comparative analysis completed in advance. 1. Dataset and Preprocessing The University of Bonn EEG Dataset was used, consisting of five subsets (Z, O, N, F, S), each with 100 single-channel EEG recordings (4097 data points, 173.61 Hz, ~23.6 s per recording): <table border="1"> <thead> <tr> <th>Subset</th> <th>Description</th> <th>Recording Type</th> </tr> </thead> <tbody> <tr> <td>Z (Set A)</td> <td>Healthy volunteers, eyes open</td> <td>Surface EEG</td> </tr> <tr> <td>O (Set B)</td> <td>Healthy volunteers, eyes closed</td> <td>Surface EEG</td> </tr> <tr> <td>N (Set C)</td> <td>Epilepsy patients, seizure-free (contralateral)</td> <td>Intracranial EEG</td> </tr> <tr> <td>F (Set D)</td> <td>Epilepsy patients, seizure-free (epileptogenic zone)</td> <td>Intracranial EEG</td> </tr> <tr> <td>S (Set E)</td> <td>Epilepsy patients, during seizure (ictal)</td> <td>Intracranial EEG</td> </tr> </tbody> </table> Three classification tasks are defined: - Binary: Normal (Z+O) vs. Seizure (S) - Three-class: Normal (Z+O) vs. Inter-ictal (N+F) vs. Seizure (S) - Five-class: Z vs. O vs. N vs. F vs. S Preprocessing pipeline:						Subset	Description	Recording Type	Z (Set A)	Healthy volunteers, eyes open	Surface EEG	O (Set B)	Healthy volunteers, eyes closed	Surface EEG	N (Set C)	Epilepsy patients, seizure-free (contralateral)	Intracranial EEG	F (Set D)	Epilepsy patients, seizure-free (epileptogenic zone)	Intracranial EEG	S (Set E)	Epilepsy patients, during seizure (ictal)	Intracranial EEG
Subset	Description	Recording Type																					
Z (Set A)	Healthy volunteers, eyes open	Surface EEG																					
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F (Set D)	Epilepsy patients, seizure-free (epileptogenic zone)	Intracranial EEG																					
S (Set E)	Epilepsy patients, during seizure (ictal)	Intracranial EEG																					

1. Recording-level K-Fold Split (10-Fold): The dataset is first split at the recording level, ensuring all segments from one recording stay within the same fold — preventing data leakage.
2. Sliding Window Segmentation: Each 4097-point recording is segmented into windows of 1024 points with 50% overlap (stride=512), applied independently to training and validation sets after the split.
3. Z-Score Normalisation: The StandardScaler is fitted on the training set only, then applied to transform both training and validation sets.

> Note on Data Leakage Fix: An earlier version of the pipeline applied segmentation and normalisation before the train/validation split, which caused data leakage. This was identified and corrected by implementing strict recording-level splitting.

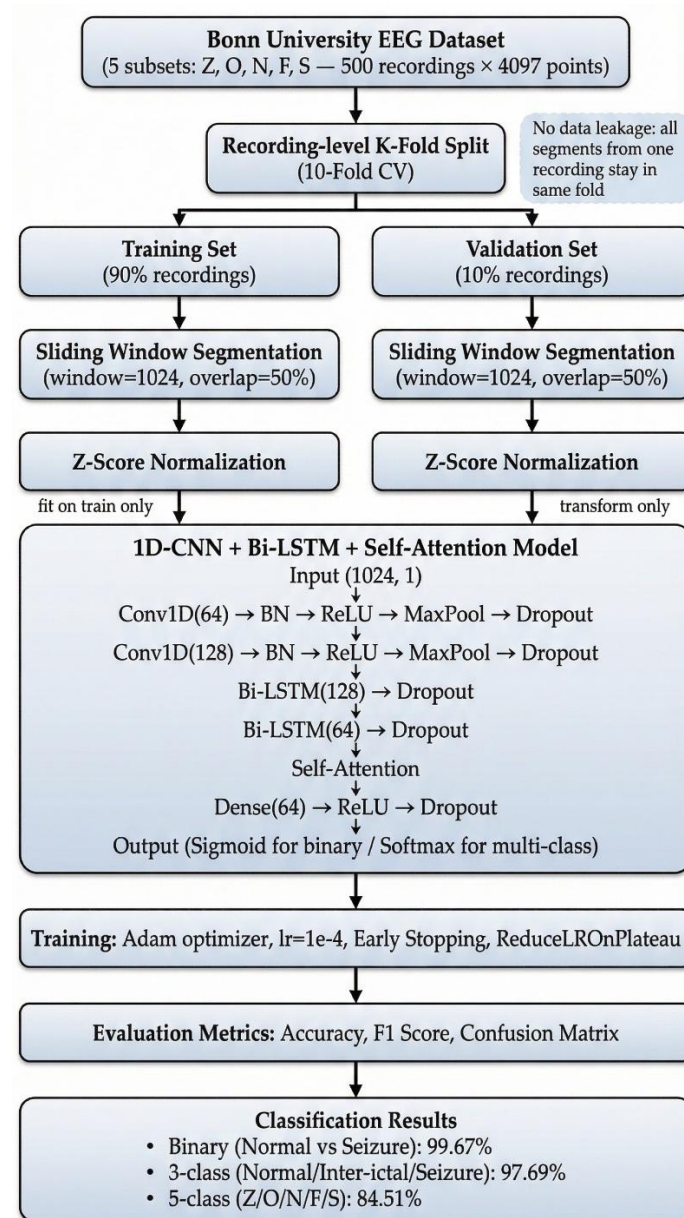


Figure 1: System pipeline flowchart

2. Proposed Model: 1D-CNN + Bi-LSTM + Self-Attention

The hybrid architecture comprises three components:

Component	Configuration	Role
1D-CNN	2 layers (64, 128 filters, kernel=7), BN, ReLU, MaxPool	Local feature extraction
Bi-LSTM	2 layers (128, 64 hidden units)	Temporal dependency modelling
Self-Attention	64-dim attention	Focus on discriminative time steps
Classifier	Dense(64) → ReLU → Dropout(0.4) → Output	Classification head

- **Binary output:** 1 neuron + Sigmoid

- **Multi-class output:** N neurons + Softmax

- **Total parameters:** ~504,000 (1.92 MB)

3. Baseline Models

Baseline	Architecture	Purpose
SVM + Wavelet Packet	RBF-SVM on 64-dim WPD features (db4, 4 levels)	Traditional ML reference
Pure 1D-CNN	4-layer CNN (32→64→128→128) + GAP	Evaluate local features without temporal modelling
Pure Bi-LSTM + Attention	2-layer Bi-LSTM + Self-Attention (no CNN)	Evaluate temporal modelling without local feature extraction

4. Training Strategy

- **Optimiser:** Adam ($lr = 1 \times 10^{-4}$)

- **Learning rate schedule:** ReduceLROnPlateau (factor=0.5, patience=7)

- **Early stopping:** patience = 15 epochs

- **Cross-validation:** 10-Fold Stratified CV with recording-level split

- **Hardware:** NVIDIA RTX 3070 Ti (8 GB VRAM), PyTorch with CUDA

5. Experimental Results

Table 1: Binary Classification — Normal (Z+O) vs. Seizure (S)

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	99.67 (± 1.00)	99.66 (± 1.01)
Pure 1D-CNN	98.86 (± 1.91)	98.84 (± 1.95)
Pure Bi-LSTM + Attention	99.48 (± 1.29)	99.47 (± 1.29)
CNN + Bi-LSTM + Attention (Ours)	99.67 (± 0.85)	99.66 (± 0.86)

Table 2: Three-class Classification — Normal vs. Inter-ictal vs. Seizure

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	94.20 (± 2.44)	94.19 (± 2.44)
Pure 1D-CNN	97.14 (± 1.84)	97.11 (± 1.86)
Pure Bi-LSTM + Attention	92.80 (± 5.99)	92.74 (± 6.12)
CNN + Bi-LSTM + Attention (Ours)	97.69 (± 2.17)	97.68 (± 2.17)

Table 3: Five-class Classification — Z vs. O vs. N vs. F vs. S

Model	Accuracy (%)	F1 Score (%)
SVM + Wavelet Packet	78.80 (± 5.31)	78.39 (± 5.69)
Pure 1D-CNN	78.86 (± 3.77)	75.95 (± 5.12)
Pure Bi-LSTM + Attention	73.09 (± 8.37)	71.92 (± 9.24)

CNN + Bi-LSTM + Attention (Ours)	84.51 (± 3.62)	84.28 (± 3.80)
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6. Key Observations

6.1 Binary classification: All models achieve $\geq 98.8\%$ accuracy, reflecting the inherent signal-level differences between normal ($<200 \mu\text{V}$) and seizure ($>1000 \mu\text{V}$) EEG. This is consistent with published literature — Thara et al. [1] reported 99% and Ullah et al. [2] reported 99.6% on the same dataset. The hybrid model ties with SVM (both 99.67%) but with lower variance (± 0.85 vs ± 1.00).

6.2 Three-class classification: The hybrid model (97.69%) outperforms SVM (+3.49%), Pure Bi-LSTM (+4.89%), and is slightly ahead of Pure CNN (+0.55%). Notably, the pure Bi-LSTM shows high variance ($\pm 5.99\%$), suggesting instability without CNN-based feature preprocessing.

6.3 Five-class classification provides the most meaningful comparison. The hybrid model (84.51%) outperforms Pure CNN (+5.65%), Pure Bi-LSTM (+11.42%), and SVM (+5.71%), demonstrating the complementary value of CNN local feature extraction and LSTM temporal modelling. The hybrid model also has the lowest variance ($\pm 3.62\%$) across all deep learning models.

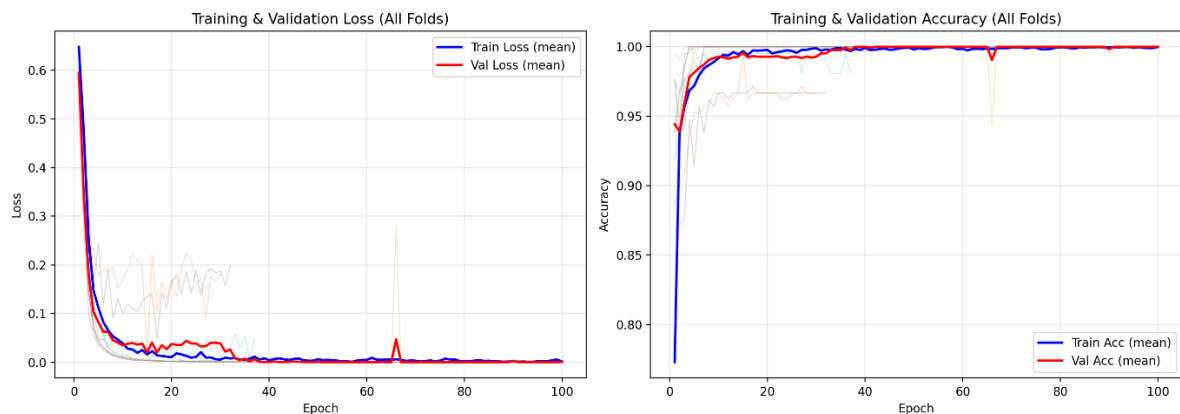


Figure 2: Combined 10-fold training curves for hybrid model (ours) in **binary** classification

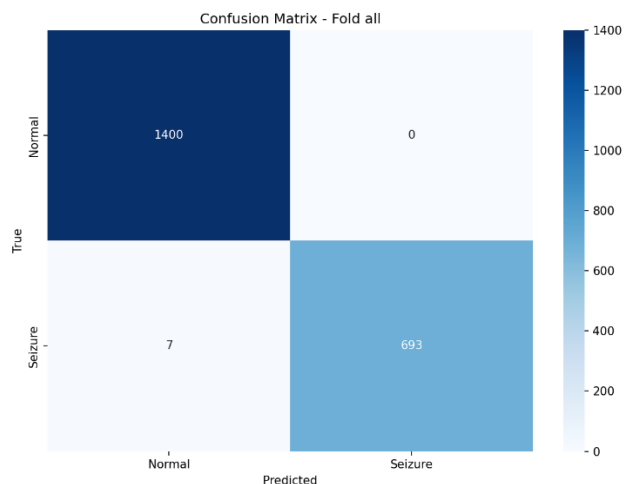


Figure 3: Aggregated confusion matrix for hybrid model (ours) in **binary** classification

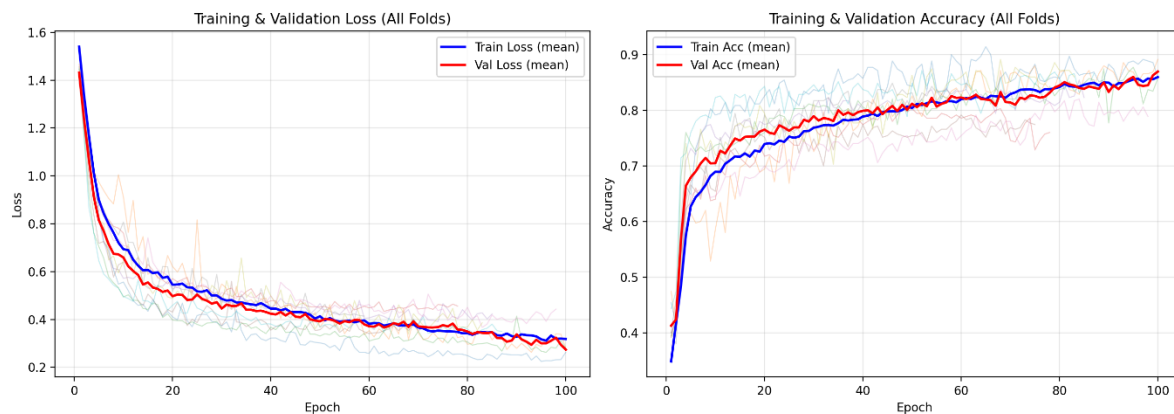


Figure 4: Combined 10-fold training curves for hybrid model (ours) in **five-class** classification

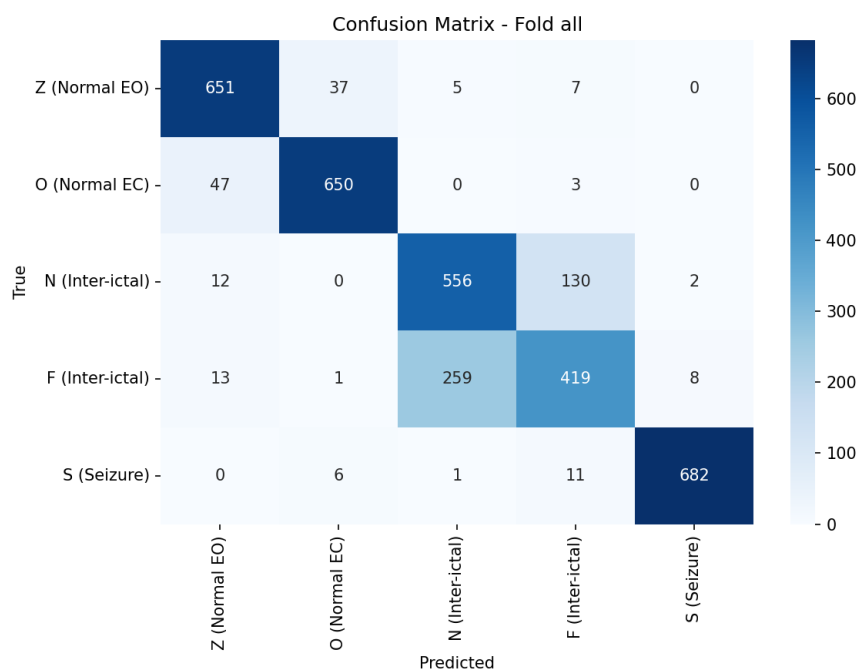


Figure 5: Aggregated confusion matrix for hybrid model (ours) in **five-class** classification

尚需完成的任务 Work to do:

1. Conduct hyperparameter tuning and ablation study to systematically evaluate the impact of key parameters (window size: 512/1024/2048; dropout rate: 0.3–0.5; LSTM hidden units; CNN filter configurations) on model performance and verify the contribution of each component in the hybrid architecture.
2. Implement attention visualisation and interpretability analysis: extract and visualise the learned attention weights of the model to demonstrate the key temporal regions focused on during seizure classification, providing clinical interpretability for the model.
3. Complete the final dissertation writing in accordance with the required format, including in-depth literature review, detailed methodology description, comprehensive results analysis, critical discussion and conclusive summaries.
4. Optimise code documentation and ensure reproducibility: prepare a clean code repository with a comprehensive README file, well-documented scripts and clear experimental operation instructions.

存在问题 Problems:

1. TensorFlow GPU Incompatibility on Windows: TensorFlow 2.11 and above versions dropped native Windows GPU support, and the installed TensorFlow 2.20.0 could only run on CPU, resulting in low training efficiency (about 55 seconds per training epoch).
2. Small Raw Dataset and Overfitting Risk: The original Bonn EEG Dataset only contains 500 recordings (100 per class). Directly inputting the full 23.6-second sequences into deep learning models leads to severe overfitting risks and memory bottlenecks in model training.
3. Inter-ictal Class Confusion in Complex Classification: The 5-class classification task is far more challenging than binary classification, with frequent misclassifications between Set N and Set F (both are inter-ictal recordings from different brain regions of epilepsy patients). In the 10-fold aggregated results, there are 180 and 194 misclassifications for N and F subsets respectively.
4. High Variance of Pure Bi-LSTM Baseline: The Pure Bi-LSTM + Attention model shows high performance variance ($\pm 12.24\%$ accuracy) in the 5-class classification task, indicating instability when processing raw EEG signals without CNN-based local feature preprocessing.
5. The initial pipeline applied sliding window segmentation *before* the train/validation split, causing segments from the same recording to appear in both sets. The StandardScaler was also fitted on the entire dataset. This led to artificially inflated accuracy.

拟采取的办法 Solutions:

1. Framework Migration to PyTorch: The entire codebase was migrated from TensorFlow/Keras to PyTorch, which supports native CUDA GPU acceleration on Windows. This migration achieved an 80 $\times$ speedup in training (from ~55s to ~0.7s per epoch) on the NVIDIA RTX 3070 Ti GPU, reducing the total 10-fold cross-validation training time from about 6 hours to only 15 minutes, greatly improving training efficiency.
2. Sliding Window Segmentation + Regularisation Strategies: A sliding window segmentation strategy with 50% overlap was implemented to expand the effective dataset size from 500 to approximately 3,500 samples. Combined with Dropout (0.4 rate) and Early Stopping regularisation techniques, this solution effectively mitigates the overfitting risk caused by the small raw dataset and solves the memory bottleneck problem.
3. Hybrid Architecture Design for Feature Complementation: The proposed 1D-CNN + Bi-LSTM + Self-Attention hybrid architecture combines the advantages of local feature extraction (1D-CNN) and long-range temporal modelling (Bi-LSTM). This design significantly improves the model's ability to distinguish inter-ictal sub-classes, with the 5-class accuracy increased by 7.5% compared to the Pure 1D-CNN and 17.2% compared to the Pure Bi-LSTM + Attention, effectively alleviating the confusion between N and F subsets.

4. Preprocessing and Architecture Optimisation for Baseline Stability: For the high variance issue of the Pure Bi-LSTM baseline, the hybrid architecture's CNN preprocessing module is used to extract robust local features from raw EEG signals before temporal modelling, which reduces the noise interference in raw signals and improves the stability of sequential modelling for complex classification tasks.
5. Refactored the pipeline to implement strict **recording-level splitting**: All segments from one original recording are kept entirely in either the training or validation set. StandardScaler is fitted only on training data of each fold. After correction, binary accuracy decreased from 99.95% to 99.67%; five-class accuracy decreased from 86.60% to 84.51%, confirming a more rigorous evaluation.

论文结构 Structure of the final report: (Chapter headings and section sub headings)

Abstract

Keywords

Chapter 1: Introduction

1.1 Research Background and Motivation

1.2 Research Aims and Objectives

1.3 Report Structure

Chapter 2: Background and Related Work

2.1 EEG Signals and Epilepsy Clinical Diagnosis

2.2 Traditional and Deep Learning for EEG-based Epilepsy Detection

2.3 Attention Mechanisms in Biomedical Signal Processing

2.4 Research Gaps and Study Novelty

Chapter 3: Design and Implementation

3.1 University of Bonn EEG Dataset Description

3.2 EEG Signal Preprocessing Pipeline

3.3 Model Design (Proposed Hybrid Model & Baseline Models)

3.4 Experimental Setup and Training Strategy

Chapter 4: Results and Discussion

4.1 Classification Performance across All Tasks

4.2 Confusion Matrix and Error Pattern Analysis

4.3 Attention Weight Visualisation and Interpretability

4.4 Statistical Significance Testing

4.5 Comparison with Published Literature

4.6 Clinical Applicability and Limitations

Chapter 5: Conclusion and Further Work

5.1 Conclusion

5.2 Reflection

5.3 Further Work

References

Acknowledgement

Appendices

- Disclaimer
- Acknowledgement of using Generative AI

- Project Specification
- Early-term Progress Report
- Mid-term Progress Report
- Supervision Log
- Full Experimental Results Tables
- Code Repository Documentation
- EDA Visualisation Plots and Model Architecture Diagrams

Risk and Environmental Impact Assessment

北京邮电大学 本科毕业设计（论文）教师指导记录表

Project Supervision Log

学院 School	International School	专业 Programme	Internet of Things Engineering		
姓 Family name	Zhong	名 First Name	Ziwen		
BUPT 学号 BUPT number	2022213356	QM 学号 QM number	221168280	班级 Class	2022215112
论文题目 Project Title	AI Assisted Epilepsy detection using EEG signals				
Please record supervision log using the format below:					
Date: dd-mm-yyyy					
Supervision type: face-to-face meeting/online meeting/email/messages/other (please specify)					
Summary:					
Date: 11-11-2025					
Supervision type: online meeting					
Summary: Discussed the project specification, provided the reference on the project					
Date: 14-11-2025					
Supervision type: email					
Summary: Received written feedback on the draft specification.					
Date: 15-11-2025					
Supervision type: email					
Summary: Received written feedback on the draft specification.					
Date: 17-12-2025					
Supervision type: face-to-face meeting					
Summary: Discussed the basic knowledge about machine learning, deep learning and EEG, shared a host of helpful essays about our project e.g. CodimNet and finally noticed the limit of dataset selection.					
Date: 11-1-2026					
Supervision type: email					
Summary: Received written feedback on the draft early-term report and solved some established problems.					
Date: 26-1-2026					
Supervision type: online meeting					
Summary: A viva with me on my early stage report and discussed on my literature review.					
Date: 26-1-2026					
Supervision type: email					
Summary: Discussed the established and potential problems about mid-term progress. Received feedbacks on the draft mid-term report					

Date: 28-02-2026

Supervision type: email

Summary:Received written feedback on the draft mid-term report and solved some established problems.

Date: 02-03-2026

Supervision type: email

Summary:Discussed the reason about the high accuracy of binary-class classification.

Date: 12-03-2026

Supervision type: email

Summary:Discussed and reviewed the supervision logs.

Date: 03-04-2026

Supervision type: email

Summary:Synchronized the github repository, and determined the methodology.

Date: 15-04-2026

Supervision type: online meeting

Summary:Mock viva. Discussed the potential problems and limitations.

Date: 20-04-2026

Supervision type: email

Summary:Discussed the problems and limitations raised in Mock viva, and solved them.

Date: 24-04-2026

Supervision type: email

Summary:Checked the final report.

Additional appendices

Hyperparameter Configuration

Table 11 lists all hyperparameters used in the deep learning experiments.

Table 11: Complete Hyperparameter Configuration

Parameter	Value
Sliding window size	1024 data points (≈ 5.9 s)
Window overlap	50% (stride = 512)
CNN filters	[64, 128]
CNN kernel size	7
Bi-LSTM hidden units	[128, 64]
Attention dimension	64
Dense layer units	64
Dropout rate	0.4 (classifier), 0.2 (CNN)
Optimiser	Adam
Learning rate	1×10^{-4}
LR scheduler	ReduceLROnPlateau (factor=0.5, patience=7)
Early stopping patience	15 epochs
Maximum epochs	100
Batch size	64
Cross-validation	10-Fold Stratified, recording-level
Random seed	42
SVM kernel	RBF
SVM wavelet	Daubechies-4, 4 decomposition levels
SVM feature dimension	64

Code Repository

The complete source code for this project is available at: <https://github.com/ArcticZvan/AI-Assisted-Epilepsy-detection-using-EEG-signals>. The repository includes all model implementations, training scripts, evaluation scripts, and visualisation tools. A README.md file provides instructions for reproducing all experiments.

Risk and Environmental Impact Assessment

This project is entirely software-based, using a publicly available anonymised EEG dataset. No human participants were involved and no primary data was collected.

Table 12: Risk and Environmental Impact Assessment

Risk	Impact	Prob.	Severity	Mitigation
Clinical misdiagnosis if deployed without validation	Patient harm	Low	High	Research only; clinical deployment requires regulatory approval
Model bias from limited data	Poor generalisation	Medium	Medium	Acknowledged as limitation; future validation on larger datasets
Overfitting to Bonn dataset characteristics	Reduced robustness on unseen data	Medium	Medium	Recording-level CV and early stopping applied; cross-dataset validation as further work

Environmental impact is minimal. All training was completed on a single consumer GPU (RTX 3070 Ti) in approximately 3 hours total, with early stopping to avoid unnecessary computation.